

**Equity Return Forecasting on Tail-Risk-Aware Portfolio Optimization on the
Nigerian Exchange Group (NGX)**

BY

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Chapter 1

1.1 Background to the Study

Capital markets play a crucial role in national economic development as they provide the mechanism for listed companies to raise capital and investors to allocate capital to optimal uses (Sulaiman, 2023; Evbuomwan, 2024). The stock market - influenced by factors such as demand, supply and investor behavior amongst others, is where listed companies raise capital by selling equity stakes through shares, while investors buy and sell these securities (Nahari, 2025). Investment in the capital markets has been on the rise as one of the many investment vehicles available to individual and institutional investors in recent times. The goal of rational investors in these markets is to minimize risk and maximize return on investments (Mahadevaswamy and Shyamala, 2022), based on their level of risk-aversion when they invest in securities. They share the same goals of diversifying all diversifiable risks and achieving the best possible risk-return ratio (Zaimovic *et al.*, 2024). This connotes the importance of diversification as a major means of sustainable capital appreciation.

Diversification is not an automatic risk-reduction mechanism as its effectiveness depends on the co-movement of the assets returns across markets and over time. When asset returns have low correlation, unsystematic risk of the portfolio can be significantly reduced through diversification. Although diversification is widely used and popular, it still presents a continually evolving challenge in portfolio management - how to optimally allocate resources. For example, international equity diversification was historically more reliable for U.S. investors prior to 2009, however, the post-crisis period favored diversification through a portfolio of stocks and bonds involving equities, commodities, and government securities. As the correlation among markets

increases over time, international portfolio diversification remains if investors continuously measure correlation among these markets so as to create an optimal market combination (Huang *et al.*, 2025).

To deal with the problems that come with diversification, portfolio management has become a step-by-step decision process. The portfolio management includes identifying, selecting and overseeing investments by individuals or institutions with the goal of maximizing returns while minimizing risk (Salo *et al.*, 2024). As markets keep evolving, portfolio management is essential for long-term growth to maximize returns (Yadav *et al.*, 2024). The main goal of portfolio construction is to diversify unsystematic risk through a two-step allocation plan. Firstly, investors balance risk-return by splitting funds between the risk-free assets and the risky assets. After the division, diversification is realized by combining assets with low correlation thereby improving the portfolio's risk-return profile and moving the efficient frontier to better results (Zhang, 2024). Within this framework, the challenge of determining a portfolio composition is an important decision problem which can be solved through portfolio optimization techniques (Salo *et al.*, 2024).

Portfolio optimization is one of the key tenets of investment theory, providing a systematic strategy for dealing with market unpredictability. This offers a systematic framework to help investors find a balance between risk and return and thus maximize their returns on every unit of risk they take. Thus, the portfolio optimization process helps in guiding investors through the uncertainty, volatility, and unpredictability of financial markets, and assists in meeting their unique financial goals (Nahari, 2025). The market downturns that have occurred, including the global financial crisis of 2008 (Global Financial Crisis), were evidence of the inadequacy of traditional

diversification methods. Consequently, there has been a greater demand for and usage of models that allow for a more sophisticated analysis of correlated risk (Senescall & Low, 2024). Optimization models provide a structure for an investor's decision-making and integrate multiple dimensions of risk, including credit, market and liquidity risk, into a single process of making investment decisions rather than relying on traditional methods of diversification (Tjiwidjaja, 2025).

The mean-variance model introduced by Harry Markowitz (1952) is one of the most foundational portfolio optimization models, which related returns and risks of an asset to its expectation and variance of returns which led to the efficient frontier concept (Zsurkis *et al.*, 2024). His foundational contributions laid the foundations for the modern portfolio theory (Mba *et al.*, 2022). By introducing mathematical optimization techniques, the asset allocation problem was formalized into an optimization framework based on the trade-off between expected return and risk. Within this framework, investors may either minimize portfolio risk for a given level of expected return or maximize expected return subject to a specified risk constraint, leading to the derivation of the efficient frontier. The theory is based on various assumptions such as: Homogeneity of investors' expectations, investors' decisions based solely on expected returns and variances, exhibiting rational risk-averse behavior. Financial markets are assumed to be frictionless and efficient with freely divisible assets, unrestricted short selling, and prices that fully reflect available information. Asset returns are uncertain, normally distributed, and correlated, while individual investors are price takers whose trades do not affect market prices (He, 2022).

While Markowitz's mean-variance portfolio theory remains foundational in finance research, it has practical limitations. One key challenge arises from errors in estimating the expected return vector

and covariance matrix, which can lead to portfolios that differ substantially from the true optimum (Job, 2022). A second limitation is that MVO does not account for tail risk. This is because of variance which measures deviation from mean and this can result in huge losses for investors during extreme events (Wang *et al.*, 2024) especially in emerging markets which are marked by high volatility, low liquidity and asymmetric information (Arif *et al.*, 2020). Variance as a risk of measure is also flawed in the sense that it treats both upside and downside deviations equally. This is against the goal of investors as upside deviations are not only welcome but desired and they are mostly concerned with downside risk (Ferrari *et al.*, 2024). This shows that the mean-variance model doesn't truly perceive the investors' perception of risk.

Due to the limitations of the mean and variance approach, other risk measures like Value-at-Risk emerged. Value-at-Risk was particularly introduced since it focuses on assessing the risk based on the highest potential loss that can occur within a specific Confidence Interval (Wang *et al.*, 2024). Value-at-Risk (VaR), a statistical risk measure that aims to quantify potential losses via a specific quantile of the conditional return distribution of a portfolio, has emerged as a common tool for market risk assessment and remains a fundamental piece of risk management practices according to the Basel III Accord (Cheng, 2025; Leccadito *et al.*, 2024).

However, VaR quickly gained acceptance as the standard approach for modeling market risk for several years because of its simplicity. Yet, VaR faces challenges that investors should be aware of. Findings from various studies have shown that during the occurrence of pandemics, VaR models were not accurate at all as far as investors were concerned, even during crises, as noted by Fatouros *et al.* (2022). Moreover, VaR models not only give data concerning potential losses at a certain confidence level, but neglect losses of greater magnitude, as pointed out by

(Mozumder et al., 2024). Moreover, it is not sub-additive, meaning the sum of individual asset VaRs can underestimate total portfolio risk. (Leccadito *et al.*, 2024). All of these highlight the need for a risk measure to show severity of extreme losses thereby showing the need for a technique that goes beyond the traditional measures thereby aiding investors in the construction of a more robust portfolio (Fan, 2020).

Consequently, portfolio optimization models based solely on historical data can misrepresent future risk, indicating the need for methods that incorporate predictive approaches (Ślusarczyk and Ślusarczyk, 2025). Empirical studies highlight the limitations of using historical data for predictive models such as unreliability during economic uncertainty or financial crises, everchanging market conditions, over fitting and bias of data mining (Zhang, 2025), ineffective capturing of nonlinear and time-varying dynamics (Chung *et al.*, 2025). Historical models have also been observed to do not sufficiently account for volatility features in emerging markets due to poor out-of-sample forecasting (Sahiner, 2022; Adegboyo, 2025). These limitations highlight the increasing difficulty of accurately forecasting risk and return in unstable and emerging markets, raising important questions about how future portfolio outcomes can be reliably anticipated.

Emerging markets assets returns and risk forecasts face distinct challenging conditions. Volatility in emerging market stocks is high, making it challenging to rely on conventional forecasting models (Alfeus et al., 2024). Conventional forecasting models cannot fit financial time series data because of the stability criteria used in the conventional model, as the financial time series data is prone to stability criteria-related problems because of randomness and randomness-related ingredients of the financial time series data, thereby emphasizing the demand for increased

precision forecasting models like Machine Learning Models, which have attracted considerable attention and used elaborately in stock market predictions and forecasting (Ashrafzadeh et al., 2025). Various empirical research works indicate the efficiency and applicability of ML Models in understanding the complex patterns and dynamics present in financial time series data, leading to increased precision and profits in stock market investments (Fan, 2020; Ashrafzadeh et al., 2025; Kevin and Yugopuspito, 2025). Hence, forecasting models not only assist in estimating predictable risks and rewards but are also helpful in understanding the importance of risk-aware optimization techniques.

Moreover, due to the various benefits in using predictive models, the forecasts can be plugged into optimization techniques - a largely untapped potential to improve outcomes (Zhang, 2024). Machine learning models improve the quality of inputs estimates which helps investors in making better portfolio decisions (Martínez-Barbero, 2024). In emerging markets marked by volatility, structural deficiencies and noisy data, ML models have been applied to stock prices forecasting, predicting market crashes, and capturing non-linear relationships in financial time series data (Naeem et al., 2024; Ashrafzadeh et al., 2025). Integrating these forecasts into portfolio optimization frameworks can improve allocation decisions and better minimize portfolio risk (Kevin & Yugopuspito, 2025; Uzoaga, 2025).

Despite several decades of progress in portfolio optimization from the mean variance model in 1952, several challenges are found within and around existing models. The conventional models for mean variance are flawed due to error in estimation with symmetry between upside and downside risks. Furthermore, Value-at-Risk models are inefficient as they are unable to determine the extent or value of tail losses and demonstrate unsteady forecasts using actual historic data for

volatile and developing markets (Job, 2022; Wang et al., 2024; Arif et al., 2020; Mozumder et al., 2024). The above factors illustrate and exemplify gaps and needs for improved portfolio optimization tools with accurate and consistent results and recommendations (Ślusarczyk and Ślusarczyk, 2025). The intersection and fusion between predictive models such as machine learning models and Tail-Risk aware models for portfolio optimization, recommend and express a positive potential for improvement within portfolio and risk management decisions in volatile markets and developing markets (Ashrafzadeh et al., 2025; Kevin and Yugopuspito, 2025), and accordingly underlay and serve as background for this research work and project.

1.2 Statement of the Research Problem

Portfolio optimization is a core mechanism in the markets for balancing risk and return. The popularity arose due to the increased size of the world economy providing investors with an ever-increasing variety of investment opportunities (Lii, 2023). This is seen on Nigerian Exchange Group (NGX) - one of Africa's leading financial hubs, where the number of securities listed has increased from 19 - at inception in 1961 to 393 listed securities among which 151 were listed companies as at September 2024 (NGX website). Portfolio optimization research remained a significant concern for researchers for efficient allocation of capital, which entails the allocation of capital based on market fluctuations (Espiga-Fernández et al., 2024). The nature of financial markets, which entails market volatility, vagueness, and market shifts, hampers the predictability of market fluctuations, as seen on the NGX. (Uzoaga et al., 2025). Nevertheless, in efforts to provide a suitable approach, there is a recent emphasis on the integration of predictive models in analytics for a better approach for non-linear analysis, and market shifts (Alfzari et al., 2025).

Whether the current approach in portfolio optimization can provide a suitable approach for market risk remains a concern.

In theory, portfolio optimization enables investors to construct portfolios that maximize expected returns for a given level of risk, or equivalently, minimize risk for a given return (EspigaFernández *et al.*, 2024). This is done by allocating capital across multiple assets, balancing correlations and individual assets risk to achieve an efficient portfolio (Zsurkis *et al.*, 2024). By combining uncorrelated assets, investors could diversify away all unsystematic risk potentially improving risk-adjusted returns (Zaimovic *et al.*, 2024). However, to achieve these theoretical benefits depends critically on accurate estimates of returns, covariance, stock forecasts and other input parameters (Zhang, 2025; Ashrafzadeh *et al.*, 2025). However, misestimation of these inputs can distort the portfolio's risk-return profile and investment outcomes (Job, 2022; Wang *et al.*, 2024; Arif *et al.*, 2020). This suggests that these advantages critically depend on how risk and returns are measured and anticipated.

However, in periods of crises, extreme events and recessions such as the 2020 Covid-19 pandemic, 2008 financial crisis, the classical optimization models proved to be inefficient (Fatouros *et al.*, 2023). Empirical studies also show that post-2009 diversification delivers less benefits as correlations among assets rise (Huang *et al.*, 2025). Traditional measures underestimate extreme losses such as Value-at-risk (VaR) which has been criticized not to show the magnitude of extreme losses and not sub-additive (Mozumder *et al.*, 2024; Leccadito *et al.*, 2024). In emerging markets, these issues are amplified due to structural issues, illiquidity, and fat tail distributions (Ojo & Okafor, 2024). Nevertheless, the persistence of these challenges has driven research into alternative risk measures and adaptive optimization methods (Lorimer *et al.*, 2024). Recent

research has attempted to address these issues through alternative risk measures and time-series forecasting models such as Expected Shortfall (ES), Autoregressive Integrated Moving Average (ARIMA) and generalized autoregressive conditional heteroskedasticity (GARCH) (Ojo & Okafor, 2024; Mozumder *et al.*, 2025). While these studies and recent research are informative, the issues don't stem from the optimization models themselves, but how expected risks and returns are computed.

Portfolio optimization failures are primarily driven by poor estimation of risk and returns. Studies have criticised the mean-variance optimization model for estimation errors in the expected return vector and covariance matrix (Job, 2022) giving rise to poor out-of-sample performance (Martínez-Barbero *et al.*, 2025). Although they have achieved significant progress, they are still insufficient as forecasts are made from historical-data performance, have stability assumptions, failure to capture non-linear and time varying dynamics (Ślusarczyk and Ślusarczyk, 2025; Chung *et al.*, 2025). This highlights the need for approaches that can better anticipate future market behavior rather than rely solely on past observations.

In empirical applications, portfolio allocations are generally quite sensitive to calculations for estimated risks and returns (Martínez-Barbero *et al.*, 2025; Ślusarczyk and Ślusarczyk, 2025). The portfolio optimization process essentially involves future expectations with respect to returns and risk because portfolio allocation weights determined by portfolio optimization are derived and based on their future expectations and behaviors and are not limited to their averages as determined before (Martínez-Barbero *et al.*, 2024). As illustrated by relevant research (Martínez-Barbero *et al.*, 2024), portfolios formed using future forecasts by LSTM models outperform those formed using historical return data. Implementing these forecasts within overall portfolio optimization

models for improved allocation and reduced portfolio risks can indeed provide benefits (Kevin & Yugopuspito, 2025; Uzoaga, 2025).

Portfolio optimization using classical frameworks has been observed to be highly vulnerable during periods of market stress. This arises because commonly used measures focus on using quantile-based losses and do not sufficiently account for the magnitude of losses in the tail of the return distribution. Conditional Value-at-Risk (CVaR), proposed by Rockafellar and Uryasev (2000), handles this problem effectively as it measures the average loss exceeding a given level of risk over a certain horizon thus suitable for dealing with extreme events (Wahid et al., 2025). With this measure of tail-risk rather than variance, it has been shown to be more robust than before when dealing with capital allocation with uncertainty. Later research has also confirmed that CVaR performs better than previous models on portfolio returns and Sharpe Ratio (Lorimer et al., 2024). Nonetheless, how effective CVaR is essentially depends on how well input values are estimated and can readily become more effective with forecasting models (Zhang, 2025). This is particularly important when dealing with emerging markets such as the Nigerian Exchange Group (NGX), where financial returns have rather high volatility, illiquidity, and fatted distributions. This indicates an important need for a hybrid combination of tail-risk-aware optimization and forward-looking forecasting of risk and return values to enhance portfolio robustness and returns when dealing with such a challenging environment.

1.3 Research Questions

The study aims to answer the following questions:

- Can machine-learning models improve the accuracy of stock return forecasts on the NSE?

- Does incorporating ML-based return forecasts into CVaR optimization improve portfolio performance?
- How does the ML-CVaR portfolio compare to traditional mean-variance and equalweighted portfolios?

1.4 Objectives of the Study

Following the research questions highlighted above, the broad objective of this study is to examine the effect of equity-return forecasting on tail-risk aware optimization on the Nigerian Exchange Group (NGX) while the specific objectives are to:

- Build ML models (Random Forest and XGBoost) to forecast stock returns.
- Integrate predicted returns into a CVaR optimization framework.
- Compare ML-CVaR portfolios to traditional mean-variance portfolio using metrics such as Sharpe ratio, Annualized returns, Sortino ratio, and maximum drawdown.
- Assess robustness of portfolios under extreme market conditions.

1.5 Justification of the Study

In modern finance, portfolio optimization has evolved to be central investment decision-making both in theory and practice (Espiga-Fernández *et al.*, 2024). As financial markets in emerging countries become more volatile, complex and uncertain, traditional models have gotten more strained and inadequate to capture the true state of the market (Adegboyo & Sarwar, 2025) especially during periods of stress. This is the cause for alarm as poor portfolio optimization can lead to poor risk allocation which has real economic consequences including huge losses, instability, and poor capital allocation. This has led to new areas of research including the increasing use of predictive and machine learning models to address non-linearity and timevarying

dynamics (Alfzari *et al.*, 2024; Kevin & Yugopuspito, 2025; Chen, 2025). Despite these advances, forecasting and optimization are usually treated separately as limited integration frameworks exist (Zhang, 2024). All these suggest a need for the reassessment of optimization frameworks and predictive models to better control risk under unstable conditions. Hence, this study will contribute to extant studies in the following ways:

One, classical optimization models such as mean-variance and VaR models are foundational and important, but they do not sufficiently capture downside risk and estimation errors especially in emerging markets characterized by illiquidity and structural break (Job, 2022; Wang *et al.*, 2024). By integrating ML models into Tail-Risk aware measures into portfolio optimization particularly in emerging markets, this may properly address these persistent limitations. Two, variance-based measures do not sufficiently capture the magnitude of extreme losses leaving portfolios exposed during market crises (Arif *et al.*, 2020; Wahid *et al.*, 2025). During events such as the 2008 financial crisis and 2020 Covid-19 crash, portfolios that were constructed with variance-based measures either greatly underperformed or made huge losses (Leccadito *et al.*, 2024). Due to these limitations, it justifies the use of Conditional Value-at-Risk (CVaR) as a tailrisk measure which explicitly accounts for extreme losses past a certain threshold, which is particularly relevant on the NGX.

Three, portfolio optimization frameworks based on historical averages do not sufficiently adequately capture the future behavior of expected returns and risks (Zhang, 2025). It has been observed that portfolios based on forecasted returns have consistently outperformed that based on historical averages (Martínez-Barbero *et al.*, 2024). To reduce the possibility of increased risk and lower returns, there is a critical need for the integration of forecasted returns into optimization

which is embedded in the study. Four, the traditional econometric forecasting models impose assumptions which are violated in the financial markets especially during structural changes, assumptions such as linearity and stationarity. This implies that the models won't be able to capture nonlinear relationships, time varying, and complex market dynamics (Lyu, 2024). Machine learning models bypass these limitations by learning complex and nonlinear patterns directly from the data without restrictive assumptions (Ashrafzadeh *et al.*, 2025). These improved forecasts directly make for better portfolio allocation decisions, as optimization outputs are sensitive to return estimates (Zhang, 2025). Consequently, these advantages are pronounced in emerging markets where return distributions are unstable and fattailed (Ojo & Okafor, 2024). However, to achieve the full benefits of ML forecasts, it needs to be effectively integrated into portfolio optimization frameworks rather than used in isolation.

Hence, this necessitates the study.

Five, most research works predominantly treat forecasting and portfolio optimization as separate tasks. This separation leads to suboptimal portfolios as optimization depends directly on forecast quality (Martínez-Barbero *et al.*, 2024). Although joint forecasting-optimization models have been observed to outperform traditional models and achieve better risk-adjusted performance (Kevin & Yugopuspito, 2025), these studies are few and limited and even fewer or non-existent in emerging markets. This uniquely positions this study as a unified decision-making framework to enhance robustness, coherence, and real-world relevance of portfolio decisions. Six, most portfolio optimization and ML-based frameworks are developed and tested on data from developed markets thus limiting their applicability in emerging markets due to higher market friction, asymmetric information and higher volatility. All these leads to fat-tailed return distribution causing variance and traditional measures to underestimate extreme losses (Mozumder *et al.*, 2024; Wang *et al.*,

2024). This also makes conventional forecasting methods inadequate. Given these conditions and the need for portfolio resilience, there is a need to examine equity-return forecasts on tail-risk aware portfolio optimization on the NGX.

Given the emerging market realities, better forecasting and risk modeling are essential due to the observed improvements over traditional measures (Chaweewanchon & Chaysiri, 2022).

Together, forecasting and tail-risk aware optimizers together form a promising avenue for more resilient portfolio outcomes. While these improved methods exist, their applications in emerging markets such as the Nigerian Exchange Group (NGX) remains under-explored (Moyoweshumba & Seitshiro, 2025). This necessitates the study not only for academics and researchers but also for real decision-making.

1.6 Scope of the Study

This study examines the effect of equity-return forecasting on tail-risk aware optimization on the Nigerian Exchange Group (NGX). Weekly price data on selected companies in the NGX Pension Index between the period of 2010 - 2025 will be utilized for the study. This study could not extend beyond the selected time frame due to unavailability of price data due to the date of companies' listings. Most available data on price data on publicly listed companies were sourced from ng.investing.com which was limited to the timeframe of the study.

1.7 Organization of the Study

This study is divided into five chapters. The first chapter of this study provides the background of the study, which will further prove the need for the study. Chapter two presents the theoretical and empirical review of relevant studies in the study's subject matter. The theoretical perspective in the second chapter will introduce the adopted models in chapter three. Data analysis and presentation will be carried out in chapter four, while the concluding comments that include

summary, conclusion, recommendations, and suggestions for further studies will be made in chapter five.

CHAPTER TWO

LITERATURE REVIEW

This research will examine the current literature regarding equity return forecasting and tail-risk aware portfolio optimization in relation to Nigerian Exchange Group (NGX). The chapter will be structured into a conceptual framework, a theoretical framework, an empirical literature review, and a summary of the gaps identified in the existing literature.

2.1 Conceptual Framework

Understanding equity return forecasting and tail-risk-aware portfolio optimization has become increasingly important in modern financial decision-making due to rising market volatility and uncertainty. Forecasting improves the accuracy of return estimates while tail-risk enhances portfolio protection against extreme losses. With the growing applications of machine learning techniques and unique characteristics of emerging markets such as the Nigerian Exchange Group (NGX), these concepts have gained increased importance in investment management.

2.1.1 Portfolio Optimization

Portfolio optimization refers to the systematic process of determining the most efficient allocation of assets in an investment portfolio to achieve a specific financial objective. Its primary objective is to construct a portfolio that either maximizes returns given a specific risk or minimizes risk given a specific return. This involves selecting and weighing assets to strike a balance between risk and return. The foundation of modern portfolio optimization was introduced by Markowitz (1952) through the mean-variance framework, which formalized the relationship between risk and return and laid the groundwork for subsequent portfolio management techniques (Salo et al., 2024).

Building from portfolio optimization, an efficient allocation principle is necessary as it focuses on how resources should be distributed across assets to achieve the best possible balance between risk and return. Achieving efficiency involves strategies such as diversification, which reduces unsystematic risk by combining assets with low correlations, and careful assessment of risk-return trade-offs, which helps investors decide how much risk they are willing to accept for potential gains. Importantly, the efficiency of a portfolio depends on accurate estimation of returns, variances, and covariances, making continuous monitoring and adjustment essential. By following these principles, investors can construct portfolios that not only optimize performance but are also more resilient to market fluctuations (Zhang, 2024; Okafor and Robertson, 2022).

As a result of modern advances in technology and market complexities - increased market volatility, large datasets and emergence of new asset classes, portfolio optimization has significantly evolved from traditional optimization methods - mean-variance model, blacklitterman model and value-at-risk (VAR) model to supplementation of traditional methods or them being replaced with more advanced analytics to better handle these challenges in real time (Hsiao, 2025).

2.1.2 Tail-Risk-Aware Portfolio Optimization

Tail risk generally refers to the risk associated with extreme events that occur in the tails of a probability distribution, representing rare but severe losses. In financial markets, return distributions often exhibit fat tails, implying that extreme losses occur more frequently than a standard normal distribution would predict (Rockafellar & Uryasev, 2002; Mozumder et al., 2024). This phenomenon is often characterized by skewness and excess kurtosis (leptokurtosis), where a high kurtosis value indicates that returns are concentrated around the mean but spread significantly

toward the tails, signaling a higher likelihood of extreme volatility and outliers (Alim et al., 2024; Bali et al., 2007).

While understanding how crucial tail-risk is, traditional methods like mean-variance model use variance which treats all deviations symmetrically, therefore ineffective in distinguishing between upside and downside risk, limiting their effectiveness in capturing extreme downside outcomes i.e a critical limitation of variance is that it penalizes positive deviations (gains) and negative deviations (losses) equally (Samaniego Alcántar, 2023). This implies that upside volatility is treated as a risk to be minimized, which can lead to suboptimal outcomes for investors who view returns above the mean as desirable rather than dangerous (Zapata Quimbayo, 2025; Lorimer et al., 2024). In contrast, downside risk measures-such as semi-variance, Value-at-Risk (VaR), and Conditional Value-at-Risk (CVaR)-are asymmetric metrics. They differentiate between "good" and "bad" volatility by focusing exclusively on returns that fall below a specific target or threshold (Ferrari et al., 2024; Zsurkis et al., 2024). By isolating the negative portion of the return distribution, downside risk aligns more closely with the investor's intuitive perception of risk as the potential for financial loss rather than mere fluctuation (Mozumder et al., 2024).

Tail risk management is necessary in volatile markets such as the Nigerian Exchange Group (NGX) due to the market's characteristic leptokurtosis and fat tails, which render symmetric measures like variance inadequate for capturing extreme downside potential (Uzoaga et al., 2025). Furthermore, the NGX exhibits significant leverage effects where negative shocks amplify volatility disproportionately compared to positive news, necessitating the adoption of asymmetric models and tail-risk-aware models such as Conditional Value-at-Risk (CVaR) for robust risk assessment during periods of structural instability (Adegboyo & Sarwar, 2025; Rockafellar & Uryasev, 2002).

2.1.3 Equity Return Forecasting

Equity return forecasting refers to the analytical process of estimating the future financial performance or price direction of stock assets to enhance decision-making in portfolio management and risk mitigation (Yu, 2024; Chao, 2024). This practice operates in contrast to the Efficient Market Hypothesis (EMH) and the Random Walk Hypothesis, which postulate that asset prices reflect all available information and fluctuate randomly, rendering prediction impossible (Manogna & Kulkarni, 2025; San, 2025). However, substantial empirical evidence challenges these hypotheses, suggesting that financial time series exhibit predictable behavioral patterns, nonlinear correlations, and volatility clustering that can be modeled to generate abnormal returns (Ashrafzadeh et al., 2025; Yu, 2024).

Equity return forecasting fundamentally transforms portfolio optimization by replacing static historical averages with dynamic, forward-looking estimates, thereby allowing optimization models to adapt to non-stationary market conditions (Manogna & Kulkarni, 2025; Yu, 2024). By integrating advanced algorithms such as Long Short-Term Memory (LSTM) networks and machine learning techniques, investors can enhance asset selection and weight allocation, effectively maximize risk-adjusted returns while mitigate the estimation errors common in traditional Mean-Variance frameworks (Ashrafzadeh et al., 2025; Fan, 2025). This predictive integration facilitates active rebalancing and improves portfolio robustness during volatile periods, yielding performance that consistently surpasses passive benchmarks like the equal-weighted or buy-and-hold strategies (Ślusarczyk & Ślepaczuk, 2025; Martínez-Barbero et al., 2025).

2.1.4 Machine Learning in Financial Forecasting

While traditional econometric models such as Autoregressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) have

historically served as the foundation for financial time series analysis, their reliance on linear assumptions often limits their effectiveness in capturing the chaotic and non-linear nature of stock market volatility (San, 2025; Ajiga et al., 2024). Recent studies suggest that financial markets exhibit complex, dynamic behaviors that rigid statistical models struggle to approximate (Zhang, 2025).

Consequently, there has been a paradigm shift toward Machine Learning (ML) and Deep Learning (DL) methodologies, which offer superior flexibility in modeling high-dimensional data and identifying latent non-linear patterns without strict distributional assumptions (Chao, 2024; Gu et al., 2019). Parallel to deep learning, tree-based methods such as Random Forest (RF) and eXtreme Gradient Boosting (XGBoost) have gained traction for their ability to handle noisy data and prevent overfitting through bagging and boosting mechanisms (Moyoweshumba & Seitshiro, 2025; Chen, 2025). The integration of these techniques into portfolio optimization models enhances asset allocation decisions by providing forward-looking returns estimates and improving the balance between risk and return in portfolio performance.

2.1.5 Nigerian Exchange Group (NGX) market characteristics

The Nigerian Exchange Group (NGX), formerly known as the Nigerian Stock Exchange (NSE), operates as a pivotal financial hub in Africa, structured to facilitate capital allocation and wealth generation through the centralized trading of equities and financial liabilities. The market is characterized by significant volatility clustering and structural breaks such as regime shifts caused by external shocks like oil price fluctuations and political instability (Adegboyo & Sarwar, 2025). A fundamental characteristic of returns on the NGX is their significant deviation from the normal distribution. A stylized fact consistent with the Efficient Market Hypothesis (EMH) critiques in emerging markets. Empirical analyses of the All-Share Index (ASI) consistently reject the

Gaussian assumption, revealing that return distributions are leptokurtic-possessing higher peaks and heavier tails (fat tails) than a normal distribution (Adegboyo & Sarwar, 2025; Uzoaga et al., 2025). This implies that extreme market events and outliers occur more frequently in the Nigerian market than standard linear models would predict (Uzoaga et al., 2025). Furthermore, NGX returns often exhibit positive skewness, indicating that the distribution of returns is not symmetric and may be influenced by occasional large positive gains, although the market remains highly volatile (Uzoaga et al., 2025).

All these pose investment challenges to investments in Nigerian equities. Historical precedents, such as the 2014 oil price crash and the COVID-19 pandemic, demonstrate how external crises can trigger severe market downturns, disrupt market capitalization, and erase investor wealth (Fapetu et al., 2021). Furthermore, the market is characterized by frequent structural breaks-abrupt shifts in market regimes caused by policy changes or economic shocks-which complicate the application of traditional linear forecasting models and necessitate adaptive risk management strategies (Adegboyo & Sarwar, 2025).

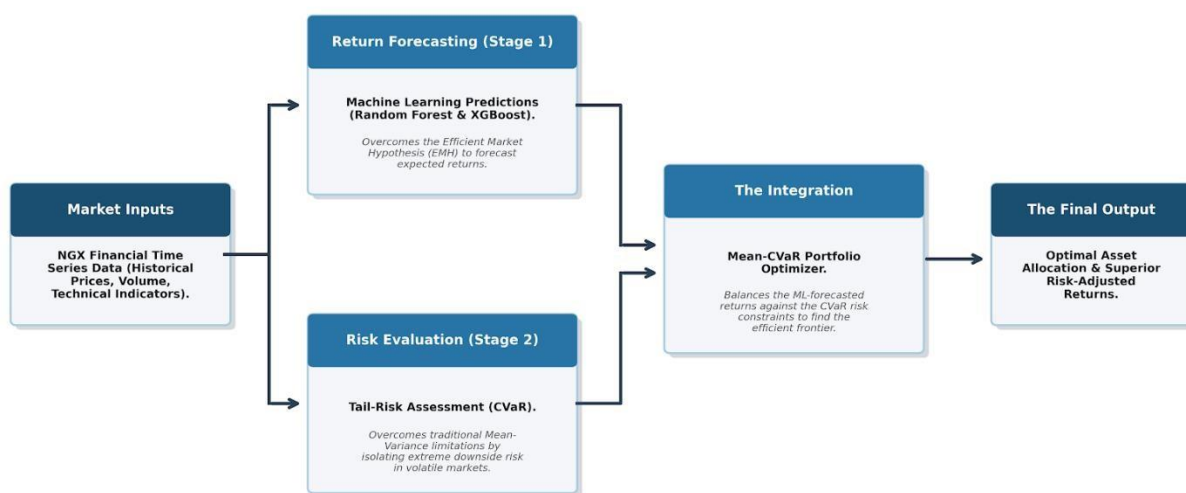


Figure 1: Diagram showing flow of conceptual framework

2.2 Theoretical Framework

This section examines economic and financial theories that may be beneficial for the study. This subsection provides the theoretical foundation for our research, which can be further utilized in our investigation.

2.2.1 Modern Portfolio Theory (MPT)

Markowitz's (1952) Modern Portfolio Theory (MPT) fundamentally posits that an asset's risk must be evaluated by its contribution to overall portfolio volatility rather than in isolation, demonstrating that diversification reduces unsystematic risk through the combination of assets with imperfect correlations (Job, 2022; Samaniego Alcántar, 2023). This optimization process defines the efficient frontier as a boundary of optimal portfolios that yields the maximum expected return for a defined level of risk or the minimum risk for a target return forming the basis for the meanvariance framework (Chaweewanchon & Chaysiri, 2022; Fan, 2025). Rational investors are thus expected to select portfolios along this frontier that align with their specific risk tolerance, avoiding inefficient allocations that offer inferior risk-return trade-offs (Bodnar et al., 2022; Zsurkis et al., 2024).

Despite its theoretical foundational status, the practical implementation of the Mean-Variance framework is severely compromised by its inherent sensitivity to estimation errors in the input parameters-specifically the expected return vector and the covariance matrix (Martínez-Barbero et al., 2025; Job, 2022). In standard practice, these parameters are unknown and are typically substituted with sample estimates derived from historical data. However, the optimization algorithm treats these estimates as true values, failing to account for parameter uncertainty (Bodnar et al., 2022; Chen, 2025). This leads to a phenomenon where the model acts as an error maximizer,

aggressively allocating capital to assets with extreme estimation errors, which results in unstable, corner-solution portfolios that lack diversification (Job, 2022; Salo et al., 2024).

Consequently, portfolios optimized under the traditional Mean-Variance model frequently exhibit poor out-of-sample performance, often failing to outperform simple heuristic benchmarks such as the naive 1/N equal-weighted strategy (Martínez-Barbero et al., 2025; Job, 2022).

2.2.2 Post Modern Portfolio Theory (PMPT)

Post-Modern Portfolio Theory (PMPT) represents a sizable shift in portfolio management by redefining risk not as general volatility, but specifically as the probability of losing money. While early theories treated all price fluctuations equally, PMPT acknowledges that rational investors do not fear unexpected gains; they only fear returns that fall below a specific target or benchmark (Alcántar, 2023). This led to the adoption of asymmetric risk measures like semi-variance, which filters out the good volatility and calculates risk based solely on returns that drop below a mean or a required rate of return (Lorimer et al., 2024). By isolating the downside, this approach aligns mathematical modeling more closely with actual investor psychology, where the pain of a loss is often felt more acutely than the pleasure of an equivalent gain (Ferrari et al., 2024).

This focus on asymmetry offers a distinct improvement over the traditional Modern Portfolio Theory (MPT). The classic MPT framework relies on the assumption that asset returns are symmetric in nature, which implies that extreme market crashes are incredibly rare events (Mozumder et al., 2024). However, empirical evidence shows that financial markets frequently exhibit fat tails and skewness, meaning that drastic losses occur far more often than MPT predicts (Adegboyo & Sarwar, 2025). By moving away from standard deviation—which penalizes upside potential just as much as downside loss—PMPT allows for the construction of portfolios that seek

to capture growth during bull markets while specifically minimizing exposure to structural breaks and crashes (Chaweewanchon & Chaysiri, 2022).

The theoretical evolution of measuring downside risk acts as a bridge between semi-variance and quantile-based metrics like Value at Risk (VaR) and Conditional Value at Risk (CVaR). While semi-variance looks at the volatility of losses, CVaR (also known as Expected Shortfall) takes this a step further by estimating the expected magnitude of a loss once a catastrophic threshold has been breached (Alotaibi et al., 2022). Unlike VaR, which simply acts as a cut-off point, CVaR is mathematically coherent, meaning it accounts for the severity of tail events and ensures that diversifying a portfolio does not theoretically increase its risk (Arif et al., 2020). Consequently, modern optimization frameworks have largely transitioned to Mean-CVaR models, which offer a more robust defense against the heavy-tailed risks inherent in volatile markets like the Nigerian Exchange Group (Quimbayo & Camacho, 2025).

2.2.3 Statistical Learning Theory

Statistical learning theory offers a mathematical basis for understanding how machine learning algorithms estimate functional relationships between variables. Unlike traditional econometrics, which often starts with a rigid, pre-defined structure (like a linear regression equation), machine learning treats the data-generating process as an unknown function that maps input variables-such as past returns or macroeconomic indicators-to an output, like future stock prices (Gu et al., 2019; Chao, 2024). The goal is to use computational power to approximate this complex function by iterating through training data, allowing the model to learn the structure of the data rather than having it imposed by the researcher (Manogna & Kulkarni, 2025). This flexibility allows these models to utilize high-dimensional datasets and extract meaningful signals from noise, a capability that is particularly valuable in financial markets where relationships are non-linear (Zhang, 2025).

A central concept in optimizing these models is the bias-variance tradeoff, which describes the tension between a model's complexity and its predictive stability. If a model is too simple, such as a basic linear regression, it tends to have high bias because it relies on strong assumptions that may not fit the actual market data, essentially missing the underlying pattern (Gu et al., 2019). On the other hand, if a model is too complex, it exhibits high variance, meaning it is so sensitive that it captures random noise and specific quirks of the training data rather than the true economic signal (Manogna & Kulkarni, 2025). The objective of statistical learning is to find the optimal balance point where the total error-comprising both bias and variance-is minimized, ensuring the model is flexible enough to capture trends but rigid enough to ignore irrelevant noise (Gu et al., 2019). Closely related to variance is the risk of overfitting, a scenario where a model performs exceptionally well on historical data-essentially memorizing it-but fails to generalize new, unseen data. In the context of financial time series, this occurs when an algorithm mistakes random market fluctuations for predictable patterns, leading to poor performance when applied to real-world trading (Ślusarczyk & Ślepaczuk, 2025; Yu, 2024). To prevent this, researchers employ regularization techniques, which mathematically penalize models for becoming too complex, effectively shrinking the estimated parameters to prevent them from reacting to noise (Gu et al., 2019; Manogna & Kulkarni, 2025). Furthermore, the generalization capability of a model is rigorously tested by splitting data into distinct training, validation, and testing sets, ensuring that the reported accuracy reflects true predictive power rather than in-sample optimization (Ashrafzadeh et al., 2025; Yu, 2024).

The advantage of machine learning lies in its universal approximation property, which allows it to model complex, non-linear data structures that rigid econometric frameworks fail to capture (Gu et al., 2019). While traditional linear models struggle with stylized facts like heavy tails, algorithms

such as XGBoost and LSTM autonomously identify latent dependencies in high-dimensional environments (Ashrafzadeh et al., 2025). This flexibility ensures robustness against shifting market regimes, effectively reducing model misspecification risks compared to static statistical approaches (Manogna & Kulkarni, 2025).

2.3 Empirical Framework

This section of the literature review offers a summary of the connection between portfolio optimization techniques, equity return forecasting and integrated forecasting approach and their individual risk-adjusted return performance.

2.3.1 Mean-Variance Portfolio Optimization

In developed markets, studies show that the MV model doesn't always outperform naive approaches such as an equally weighted portfolio in terms of risk-adjusted returns (like Sharpe ratio). In the US equity market, the naive approach is quite competitive and, in some cases, outperformed where the number of assets is large and unsystematic risk is low (Salo et al., 2024). Additionally, in other studies where unsystematic risk was higher, it is observed to achieve superior returns even during bearish markets showing the mean variance portfolio outshined naive approach due to proper diversification of unsystematic risk (Martínez-Barbero et al., 2025; Ślusarczyk & Ślepaczuk, 2025).

In emerging markets, the MV model didn't perform as well due to issues like high volatility and low liquidity. According to (Job, 2022, using the MSCI Emerging and Frontier Market Index, show that the index outperformed optimization models underscoring the need for better downside risk management and the importance of emerging markets in the world economy.

The failure of the MV model in emerging markets stems from its extreme sensitivity to estimation errors in the input parameters and out-of-sample instability (Bodnar et al., 2022; Rigamonti and

Lučivjanská, 2021). Consequently, the reliance on historical sample means and covariances without robustification renders the traditional MV weak and inferior to naive approaches in practical applications (Martínez-Barbero et al., 2025). Based on these studies, it shows a need for more adaptive and downside risk-aware techniques for portfolio optimization.

2.3.2 VaR and CVaR Based Portfolio Models

Value at Risk (VaR) has become a widely accepted benchmark for measuring financial risk, endorsed by the financial regulatory body of Basel Committee on Banking Supervision (BCBS) when it was made a supplement to credit risk measure in the Basel II Accord. VaR is a measure used to determine downside risk when it was initially designed to be used for risk assessment in financial institutions (Arif et al., 2020; Alotaibi et al., 2022; Fatouros et al., 2023).

However, even though it has been used for decades as the standard for measuring tail risk, it lacked efficiency during economic crisis such as the 2020 Covid-19 pandemic and the 2008 global financial crisis. This can be traced to its simplistic assumptions that are summed in the phrase by Einhorn and Brown in 2008, where VaR is compared to "an airbag that works all the time, except when you have a car accident" (Fatouros et al., 2023). VaR faces major challenges such as subadditivity, inadequate percentile of distribution tail loss amongst others (Mozumder et al., 2024; Alotaibi et al., 2022).

Conditional Value at Risk (CVaR) on the other hand acts as a step up since it is derived from VaR but improves coherence properties and satisfies the shortcomings of VaR like subadditivity and convexity properties enabling it to better handle tail risk and achieving a higher accumulated risk-adjusted returns (Yan et al., 2020). According to Alotaibi et al. (2022), CVaR achieves superior returns compared to VaR underscoring a more accurate measurement of financial risk.

2.3.3 Integrated Forecasting and Optimization Models

Recent empirical studies have consistently shown that machine learning (ML) models tend to outperform traditional econometric techniques when it comes to forecasting stock returns.

Traditional models, such as ARIMA, rely heavily on the assumption that financial data is stable and linear, which often leads to errors because real-world markets are volatile and complex (Ashrafzadeh et al., 2025). In contrast, advanced algorithms like Random Forest and Neural Networks are designed to identify intricate, non-linear patterns that standard models miss (Gu et al., 2019). For example, San (2025) found that Random Forest models achieved significantly lower error rates (RMSE) compared to the hybrid ARIMA-GARCH model because they could handle market noise and outliers much better. Similarly, Martínez-Barbero et al. (2025) demonstrated that Long Short-Term Memory (LSTM) networks provided much more accurate predictions than traditional statistical methods, both in estimating price values and predicting the direction of the market.

Building superior forecasting, researchers have developed hybrid frameworks that use these machine learning predictions directly into portfolio optimization models. Instead of using historical averages-which assume the future will look like the past-these frameworks use the predicted returns generated by ML models as the inputs for the optimization process (Ashrafzadeh et al., 2025). This method allows the portfolio to adjust dynamically to new information. For instance, Chaweewanchon and Chaysiri (2022) proposed a two-step approach where stocks are first pre-selected based on high predicted returns from a neural network, and then the Markowitz model is used to assign weights. Fan (2025) and Manogna and Kulkarni (2025) utilized similar frameworks, showing that combining LSTM or XGBoost predictions with mean-variance

optimization greatly improves how portfolios are constructed by making them more responsive to market changes.

When these forecast-driven frameworks are put to the test, the economic gains are often substantial compared to passive strategies. Gu et al. (2019) provided striking evidence showing that an investor using neural network forecasts to time the S&P 500 could increase their annualized Sharpe ratio from 0.51 (buy-and-hold) to 0.77, essentially earning much higher risk-adjusted returns. This outperformance holds true even during difficult market periods; Martínez-Barbero et al. (2025) reported that prediction-based portfolios consistently beat the EURO STOXX 50 index, generating positive results even during bear markets. Furthermore, Ślusarczyk and Ślepaczuk (2025) found that strategies utilizing XGBoost forecasts outperformed standard benchmarks in terms of the Information Ratio, confirming that active management based on algorithmic predictions leads to more efficient capital allocation than simple diversification.

In the context of emerging markets like Nigeria, empirical evidence strongly supports the adoption of these hybrid machine learning approaches. Uzoaga et al. (2025) analyzed the Nigerian Stock Exchange and concluded that Random Forest techniques significantly outperformed linear regression in predicting stock prices, suggesting these tools are well-suited for the Nigerian market structure. Further comparative analysis across Africa by Uzoaga et al. (2025) revealed that while Random Forest was dominant for Nigerian stock returns, different models like Neural Networks worked better for other markets like Uganda, highlighting the need for model selection.

Additionally, a study on the Johannesburg Stock Exchange by Moyoweshumba and Seitshiro (2025) demonstrated that portfolios optimized using XGBoost and Random Forest achieved Sharpe ratios of 1.44 and 1.30 respectively, far surpassing the 0.41 ratio of the traditional

Markowitz model. This confirms that hybrid ML strategies are not just theoretical but offer practical, superior performance in African emerging economies.

2.4 Research Gap

Despite the growing body of research on portfolio optimization and equity return forecasting, significant gaps remain, particularly in the area of emerging markets like Nigeria. Most of the current studies are focused on developed economies with stable and liquid markets, which often do not capture the volatility, thin trading, and structural shocks characteristic of local markets. This focus limits understanding of how traditional models, such as mean-variance optimization, or even more advanced techniques, behave under conditions of high market uncertainty, leptokurtic returns, and sudden economic or policy shifts.

Furthermore, there is a clear shortage of research integrating predictive modeling with risk-aware portfolio construction. While machine learning and other advanced forecasting techniques have demonstrated strong predictive power, few studies explore how these forecasts can be directly used in optimizing portfolios that account for tail risk. Most research treats forecasting and optimization as separate processes, which reduces the level of application of these methods in real-world decision-making where investors need both accurate predictions and robust allocation strategies simultaneously.

Finally, the practical implementation of these models remains underexplored. Many studies rely on historical data and theoretical assumptions, without testing how forecast-driven, tail-risk-aware portfolios perform under real-world constraints such as transaction costs, liquidity limitations, and market microstructure effects. This leaves a critical gap between theory and practice, creating uncertainty about whether these advanced approaches can truly deliver superior risk-adjusted

returns in emerging markets. Addressing these gaps is essential for developing adaptive, resilient, and practically implementable investment strategies tailored to markets like the Nigerian Exchange Group.

2.5 Summary of Gaps in Literature

Overall, the literature shows a lack of research in emerging African markets especially Nigeria that bridges advanced forecasting techniques with robust risk management frameworks. There is limited integration of machine learning predictions into tail-risk-aware portfolio construction, and few studies have tested these methods in real emerging markets. This leaves important questions unanswered about the effectiveness, adaptability, and practical value of such hybrid approaches for investors navigating volatile and complex market environments. These gaps justify the need for a study that develops and evaluates a framework combining predictive forecasting with Conditional Value-at-Risk optimization tailored to Nigerian equities.

CHAPTER THREE

RESEARCH METHODOLOGY

This chapter describes the research methodology employed in the analysis of this study. It entails the theoretical framework, the model specification, the estimation techniques, measurement and description of the variables studied, and lastly, the sources of data.

This chapter is organized as follows. Section 3.1 explains the theoretical framework guiding the conduct of the study, and this is derived from the economic theories specified in Chapter Two. Section 3.2 reveals the model specification used to achieve the objectives of this study, which is in line with the theoretical framework. Section 3.3 comprises the estimation techniques used in the

study to achieve objectives. Section 3.4 provides the description and measurement of the study variables. Section 3.5 which concludes the chapter, explains how data will be obtained.

3.1 THEORETICAL FRAMEWORK

3.1.1 Modern Portfolio Theory and Tail Risk

As established in Chapter 2, the foundation of portfolio selection lies in Harry Markowitz's Modern Portfolio Theory (MPT), which formalizes the risk-return trade-off by assuming normally distributed returns and relying on symmetric variance as the primary measure of risk (Markowitz, 1952).

However, the empirical reality of emerging markets like the Nigerian Exchange Group (NGX) fundamentally violates these strict assumptions; these markets are instead characterized by high volatility, low liquidity, asymmetric information, and leptokurtic distributions where extreme downside market events happen much more frequently than normal distributions would predict (Arif & Sohail, 2020). To address these structural violations, Post-Modern Portfolio Theory (PMPT) offers a theoretical solution by redefining risk not as general symmetrical volatility, but specifically as the probability of financial loss, thereby distinguishing between desirable upside gains and detrimental downside drops (Alotaibi et al., 2022). Consequently, this methodology operationalizes PMPT by discarding the traditional mean-variance framework in favor of Conditional Value-at-Risk (CVaR), a coherent risk measure selected to specifically quantify and minimize the average severity of extreme tail losses occurring beyond a catastrophic threshold in the NGX (Rockafellar & Uryasev, 2002).

3.1.2 Machine Learning Theories for Forecasting

To address the limitations of the Efficient Market Hypothesis (EMH) outlined in Chapter 2, this methodology utilizes advanced statistical learning theories. The EMH and the associated Random Walk Hypothesis postulate that asset prices instantaneously reflect all available information and fluctuate randomly, theoretically rendering future price movements unpredictable and active forecasting futile (Fama, 1970).

However, substantial empirical evidence challenges these strict assumptions, particularly in emerging markets like the Nigerian Exchange Group (NGX), which exhibit market frictions, significant volatility clustering, frequent structural breaks, and complex non-linear correlations (Adegboyo & Sarwar, 2025). Statistical learning theories offer a robust mathematical framework that bypasses rigid econometric assumptions, allowing for the extraction of meaningful, latent non-linear dependencies directly from noisy and high-dimensional financial data (Gu, Kelly, & Xiu, 2019).

Consequently, this methodology adopts advanced ensemble machine learning algorithms—specifically Random Forest and eXtreme Gradient Boosting (XGBoost)—to effectively operationalize statistical learning, thereby capturing the complex, non-stationary market patterns that traditional models and the EMH fail to identify (Manogna & Kulkarni, 2025; Ashrafzadeh et al., 2025).

3.2 Model Specification

3.2.1 Predictive Models

The aim of the predictive phase of the study is to forecast future returns of the component assets within the NGX Pension Index. To achieve this, the study frames return forecasting as a supervised multivariate time-series problem. Unlike traditional linear asset pricing models which assume a linear relationship between risk factors and returns, this methodology posits that financial time series are driven by complex, non-linear, and interacting market dynamics.

Let R_{t+1} represent the continuous logarithmic return of a given equity asset for the upcoming week, calculated from the weekly closing prices. The foundational predictive relationship is defined as:

$$R_{t+1} = F(\mathbf{X}_t) + \varepsilon_{t+1}$$

Where:

- R_{t+1} is the dependent variable representing the weekly logarithmic return at time $t + 1$.
- F represents the complex, non-linear mapping function that will be empirically approximated by the ensemble machine learning algorithms (Random Forest and XGBoost).
- $\mathbf{X}_t$ is the multi-dimensional feature vector containing the independent variables strictly observed at the end of week t .
- ε_{t+1} is the idiosyncratic error term (or white noise) representing unobservable market shocks that occur at time $t + 1$, which are independent of the information set at time t .

To provide the machine learning algorithms with a comprehensive representation of market conditions, the feature space is populated with thirteen explicitly selected variables. These variables are derived entirely from the historical Open, High, Low, Close, and Volume (OHLCV) data to prevent multicollinearity while capturing diverse market dimensions.

$$x_t = [x_{1,t}, x_{2,t}, \dots, x_{13,t}]^T = [MACD_t, PPO_t, RSI_t, STOCH_t, R_{t-1}, R_{t-2}, R_{t-3}, R_{t-4}, SMA_t, ADX_t, SAR_t, ATR_t, OBV_t]^T$$

The feature vector x_t is mathematically defined as a 13×1 column vector:

$$X_t = \begin{bmatrix} MACD_t \\ PPO_t \\ ATR_t \\ RSI_t \\ x_{12,t}, STOCH_{SMA_t} \\ R_t \\ R_{t-1} \\ R_{t-2} \\ R_{t-3} \\ ADX_t \\ SAR_t \\ OBV_t \end{bmatrix}$$

The variables within X_t are categorized into four distinct market dimensions:

1. **Momentum and Oscillator Indicators:** The Moving Average Convergence Divergence

($MACD_t$), Percentage Price Oscillator (PPO_t), Relative Strength Index(RSI_t), and Stochastic Oscillator ($STOCH_t$) measure the velocity and magnitude of recent price swings to identify overbought or oversold conditions.

2. **Autoregressive (Lagged) Returns:** The sequence of R_t , R_{t-1} , R_{t-2} , and R_{t-3} represents the preceding four weeks of logarithmic returns, feeding historical price momentum directly into the model.

3. **Trend and Volatility Indicators:** The Simple Moving Average (SMA_t), Average Directional Index (ADX_t), and Stop and Reverse (SAR_t) capture the absolute strength and direction of the trend. Crucially for the subsequent tail-risk optimization, the Average True Range (ATR_t) captures intra-week market volatility and extreme price fluctuations.

4. **Volume Flow:** On-Balance Volume (OBV_t) acts as a proxy for market liquidity and institutional participation, validating the strength of the observed price trends.

This specific autoregressive functional form is strictly structured to avoid look-ahead bias. By ensuring that the predictive function (.) maps only the information matrix available at time t to the target return at time $t + 1$, the chronological integrity of the predictive pipeline is preserved, rendering the model suitable for practical portfolio allocation on the Nigerian Exchange Group.

3.2.2 Random Forest Specification

To capture the non-linear interactions among the thirteen technical and momentum indicators, this study employs the Random Forest (RF) algorithm, a robust ensemble machine learning technique.

1. Bootstrap Aggregating (Bagging)

The foundation of the RF model relies on the statistical method of bagging. Given the training dataset of weekly returns and features, the algorithm generates B independent bootstrap samples (random samples drawn with replacement). Let the original training dataset be defined as

$$S = \{(X_1, R_2), (X_2, R_3), \dots, (X_T, R_{T+1})\}.$$

For each iteration $b = 1, 2, \dots, B$:

A bootstrap sample S_b is drawn from S . A distinct regression tree, denoted as T_b , is grown using this specific localized sample S_b .

2. Tree Construction and Feature Randomness

At each node of the tree T_b , a random subset of features is selected from the total $p = 13$ available indicators in the feature vector X_t . For regression tasks, this subset is typically established as $m \approx p/3$.

$$\min_{\{j,s\}} \left[\sum_{x_i \in R_1(j,s)} (y_i - \hat{y}_{R1})^2 + \sum_{x_i \in R_2(j,s)} (y_i - \hat{y}_{R2})^2 \right]$$

The algorithm determines the optimal splitting variable j (from the m features) and split point s by minimizing the Mean Squared Error (MSE) across the resulting regions $R_1(j,s)$ and $R_2(j,s)$:

$$\min_{j,s} = \left[\sum_{x_i \in R_{left}(j,s)} (y_i - \hat{y}_{left})^2 + \sum_{x_i \in R_{right}(j,s)} (y_i - \hat{y}_{right})^2 \right]$$

(Where R_{left} and R_{right} are the partitioned feature spaces and $\hat{y}$ represents the mean of the observed returns within that specific region.

3. The Ensemble Forecast

Each individual tree T_b is grown deeply without pruning. While a single unpruned tree captures highly complex, non-linear market relationships with low bias, it suffers high variance (overfitting). The Random Forest model neutralizes this variance by averaging the continuous predictive outputs across all independently trained trees.

The aggregate Random Forest prediction for the weekly return at time $t + 1$, given the specific market information set $\mathbf{X}_t$, is mathematically specified as:

$$R_{tRF+1} = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{X}_t)$$

Where:

- R_{tRF+1} is the forecasted logarithmic return of the target asset.
- B is the total number of regression trees in the forest ensemble (a hyperparameter that will be optimized during the model training phase).
- $T_b(\mathbf{X}_t)$ is the predicted return of the b -th individual regression tree based on the 13-dimensional feature vector $\mathbf{X}_t$.

3.2.3 eXtreme Gradient Boosting (XGBoost)

XGBoost operates as an additive ensemble of classification and regression trees (CART), optimizing a regularized objective function via second-order gradient descent.

1. The Additive Model Specification

The XGBoost algorithm models the expected return at time $t + 1$ as an additive combination of sequential base learners (regression trees). For a given feature vector $\mathbf{X}_t$ containing the thirteen extracted market indicators, the aggregated XGBoost forecast is defined as:

$$\hat{R}_{t+1}^{XGB} = \sum_{k=1}^K f_k(\mathbf{X}_t), \quad f_k \in F$$

Where:

- $\hat{R}_{t+1}^{XGB}$ is the forecasted logarithmic return.
- K is the total number of additive trees.
- f_k represents an independent classification and regression tree (CART) structure within the functional space F of all possible trees.

2. The Regularized Objective Function

Financial time series are notoriously noisy, making complex models highly susceptible to overfitting. XGBoost addresses this by optimizing a regularized objective function at each sequential step (not to be confused with time index t). The algorithm seeks to minimize the following loss function:

$$L = \sum_{i=1}^N l(R_{i,t+1}, \hat{R}_{i,t+1}) + \sum_{k=1}^K \Omega(f_k)$$

Where:

- $l(\cdot)$ is a differentiable convex loss function, specifically the Mean Squared Error (MSE), which measures the difference between the actual observed return $R_{i,t+1}$ and the model's prediction $\hat{R}_{i,t+1}$.
- $\Omega(f_k)$ is the explicit regularization penalty applied to the model's complexity to prevent overfitting the technical indicators.

The regularization term is mathematically defined as:

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda |w|^2$$

Where T is the number of terminal nodes (leaves) in the tree, w represents the leaf weights, γ penalizes the addition of new splits (controlling tree depth), and λ is the L2 regularization term on the weights. By aggressively penalizing complexity, the XGBoost model isolates true predictive signals from transient market noise on the NGX.

3.2.4 The Tail-Risk Aware (Mean-CVaR) Optimization Model

The continuous return forecasts generated by the Random Forest and XGBoost algorithms (R^{RF} and R^{XGB}) are aggregated into an expected return vector, denoted as $\hat{\mu}$, which serves as the primary input for the optimization algorithm.

To account for the severe downside volatility inherent in emerging equity markets, this study discards the traditional Mean-Variance framework (which assumes normally distributed returns) in favor of the Mean-Conditional Value at Risk (Mean-CVaR) framework.

1. Formulation of Portfolio Loss and VaR

Let w represent the vector of portfolio weights allocated to the N constituent assets of the NGX Pension Index and r represent the vector of actual asset returns. The portfolio loss function is defined as the negative of the portfolio return:

$$L(w, r) = -w^T r$$

Before defining CVaR, the Value at Risk (VaR) at a specified confidence level $\alpha \in (0,1)$ (e.g., 95%) must be established. VaR represents the minimum loss expected in the $(1 - \alpha)$ worst-case scenarios:

$$Va(w) = \min \{ \gamma \in \mathbb{R} : P(L(w, r) \leq \gamma) \geq \alpha \}$$

2. Conditional Value at Risk (CVaR)

$$CVaR_\alpha(w) = E[L(w, r) \mid L(w, r) \geq VaR_\alpha(w)]$$

$$F_\alpha(w, \gamma) = \gamma + (1 / (1 - \alpha)) E[\max(0, -w^T R - \gamma)]$$

Following the formulation by Rockafellar and Uryasev (2000), CVaR can be seamlessly integrated into a linear programming problem by minimizing an auxiliary continuous and convex function $F_\alpha(w, \gamma)$:

$$(w, \gamma) = \gamma + \frac{1}{1 - \alpha} E[\max(0, L(w, r) - \gamma)]$$

3. The Optimization Objective

$$\min_w F_\alpha(w, \gamma) \quad \text{s.t.} \quad w^T E[R] \geq r_{\text{target}}, \quad \sum_{i=1}^N w_i = 1, \quad w_i \geq 0$$

The ultimate objective of the model is to minimize the tail risk (CVaR) while achieving a minimum acceptable expected return driven by the machine learning forecasts. The optimization problem is specified as:

$$\min_{w, \gamma} \left(\gamma + \frac{1}{q} (1 - \alpha) \sum_{j=1}^q [-w^T r_j - \gamma]^+ \right)$$

Subject to the following empirical constraints:

- **Target Return Constraint:** $w^T \hat{\mu} \geq \mu_{target}$ (The portfolio's expected return, leveraging the ML vector $\hat{\mu}$, must meet or exceed a specified target).
- **Budget Constraint:** $\sum_{i=1}^N w_i = 1$ (The portfolio is fully invested, utilizing 100% of available capital).
- **Long-Only Constraint:** $w_i \geq 0 \quad \forall i$ (Short selling is prohibited, reflecting institutional realities on the NGX).

By directly feeding the non-linear machine learning predictions ($\hat{\mu}$) into this coherent riskoptimization framework, the final portfolio captures both predictive momentum and robust defense against extreme market downturns.

3.3 Estimation Techniques

The estimation techniques applied in this study are carefully chosen to meet the stated objectives and the nature of data employed. Since the study employs time-series over the period of 2010-2025,

it is important that the employed techniques can pick up the statistical features of the data and the specific research questions

3.3.1 Time-Series Data Partitioning (Train-Test Split)

Evaluating machine learning models requires testing them on unseen, out-of-sample data. While traditional machine learning relies on randomized K-fold cross-validation, applying random splits to financial time-series data introduces *look-ahead bias*. This severe methodological flaw occurs when future market information inadvertently leaks into the training dataset, destroying the chronological order of events, and artificially inflating the model's predictive accuracy.

To preserve temporal dependencies, this study uses Rolling Window Walk-Forward Validation. This technique simulates real-world forecasting by sequentially advancing a fixed-size training window through the historical NGX dataset.

The algorithm trains the Random Forest and XGBoost models on a fixed historical lookback period of size W (e.g., the preceding 156 weeks) to forecast the return for the immediate subsequent week (t). Once the out-of-sample prediction is recorded, the entire training window shifts forward by one week. The oldest weekly observation is discarded, the most recent observation is absorbed, and the models are retrained.

3.3.2 Model Training and Hyperparameter Tuning

Out-of-the-box machine learning algorithms rarely capture the complex dynamics of financial time series optimally. Therefore, this study employs a Randomized Search Cross-Validation strategy within the training window to efficiently identify the optimal hyperparameter configurations for both models.

The tuning process focuses on penalizing model complexity to prevent overfitting:

- **Random Forest Parameters:** The search optimizes the number of trees in the forest ($n_estimators$), the maximum depth of each tree (max_depth), and the minimum number of samples required to split an internal node ($min_samples_split$).
- **XGBoost Parameters:** The search optimizes the learning rate or step size shrinkage ($learning_rate$), the maximum tree depth (max_depth), and the minimum loss reduction required to make a further partition ($gamma$), which directly controls the structural regularization.

3.3.3 Portfolio Turnover and Transaction Costs

To ensure that the optimized Mean-CVaR portfolio strategy is economically viable and not merely theoretically profitable, it is imperative to account for real-world trading frictions, such as brokerage fees and slippage on the Nigerian Exchange Group (NGX) (Ashrafzadeh et al., 2025). This requires measuring the portfolio turnover, which mathematically quantifies the absolute change in the portfolio's asset weights from one weekly rebalancing period to the next (Zsurkis, Nicolau, & Rodrigues, 2024):

Following standard empirical finance procedures, transaction costs are deducted from gross returns to construct net returns. This study evaluates portfolio performance across three execution-cost regimes: a gross frictionless baseline (0.00%), an institutional PFA execution scenario applying a fixed 75 basis points (0.75%) per rebalancing trade, and a retail execution friction scenario applying 150 basis points (1.50%) to account for brokerage commissions and market slippage on the NGX. Finally, a Net Returns feature column will be constructed by deducting these calculated transaction costs from the gross portfolio returns, providing a rigorous and realistic evaluation of the portfolio's actual out-of-sample performance.

$$\text{Turnover}_t = \sum_{i=1..N} |w_{\{t,i\}} - w_{\{t^{\wedge},i\}}|$$

3.3.4 Feature Importance Extraction

Due to the criticism of ensemble models of being black boxes in traditional econometrics and to ensure economic interpretability, this study extracts the built-in feature importance scores from

the optimized models to determine which of the thirteen technical indicators exert the strongest predictive influence on NGX equity returns.

- **Random Forest Interpretation:** Importance is calculated using the Mean Decrease in Impurity (Gini Importance). It measures the total reduction of the Mean Squared Error (MSE) brought by that specific feature across all trees in the forest.
- **XGBoost Interpretation:** Importance is measured using Information Gain. It evaluates the relative contribution of each feature to the model by calculating the improvement in accuracy it provides to the branches it is on.

3.3.5 Evaluation Metrics for Forecasting and Portfolio Performance

The methodology utilizes a dual-evaluation framework to assess both the predictive accuracy of the machine learning algorithms (Stage 1) and the financial viability of the resulting Mean-CVaR optimized portfolio against the Mean-Variance baseline (Stage 2).

1. Predictive Forecasting Metrics

To evaluate the machine learning phase, the deviation of the forecasted expected returns ($\hat{\mu}$) from the actual observed returns (R) is measured using two continuous error metrics:

- **Root Mean Squared Error (RMSE):** Penalizes larger forecasting errors more heavily, which is crucial for identifying severe prediction failures before they are fed into the optimizer.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{\mu}_i - R_i)^2}$$

$$MAE = (1 / n) \sum_{i=1}^n |y_i - \hat{y}_i|$$

- Mean Absolute Error (MAE): Measures the average absolute magnitude of the prediction errors, providing a baseline view of model accuracy without directional bias:

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{\mu}_i - R_i|$$

2. Risk-Adjusted Portfolio Metrics

To evaluate the final portfolio allocation, performance is measured using risk-adjusted return metrics. This requires a localized risk-free rate proxy (R_f), for which the de-annualized Central Bank of Nigeria (CBN) 91-day Treasury Bill rate is utilized.

$$\text{Sharpe Ratio} = (E[R_p] - R_f) / \sigma_p$$

- Sharpe Ratio: Measures the excess return per unit of total portfolio risk (standard deviation σ_p):

$$\text{Sharpe} = \frac{E[R_p] - R_f}{\sigma_p}$$

- **Sortino Ratio:** Aligning with the thesis's focus on tail risk, this ratio isolates downside volatility. It replaces total standard deviation with downside deviation (σ_d), penalizing only the returns that fall below R_f .

$$\text{Sortino} = \frac{E[R_p] - R_f}{\sigma_d}$$

- **Maximum Drawdown (MDD):** Quantifies the portfolio's absolute worst-case scenario. It measures the maximum observed loss from a historical peak to a subsequent trough before a new peak is attained.

$$MDD = \frac{Trough\ Value - Peak\ Value}{Peak\ Value}$$

By utilizing these metrics, the study empirically validates whether the non-linear machine learning forecasts, combined with CVaR tail-risk constraints, genuinely deliver superior risk-adjusted performance on the Nigerian Exchange.

3. Benchmark Portfolios for Evaluation

To rigorously evaluate the economic value and true out-of-sample performance of the advanced ML-CVaR optimization strategy, it is essential to measure its results against standard baseline portfolios. By utilizing these benchmarks alongside the metrics above, the study empirically validates whether the non-linear machine learning forecasts, combined with CVaR tail-risk constraints, genuinely deliver superior risk-adjusted performance on the Nigerian Exchange.

- **The 1/N Equally Weighted Portfolio (EWP):** In this naive diversification strategy, investment capital is simply divided uniformly across all selected stocks without relying on any parameter estimation or complex optimization algorithms. In an EWP strategy, each asset in the portfolio holds an equal weight of $w = 1/N$. This rule is chosen as a primary benchmark because it is easy to implement and continues to be a standard, highly effective allocation rule utilized by investors.
- **The Buy-and-Hold Market Index Strategy:** A passive strategy where capital is initialized at equal weights $w_{\{0,i\}} = 1/N$ at $t = 0$. Post-initialization, asset weights drift passively with market price returns $(w_{\{t+1,i\}} \propto w_{\{t,i\}}(1 + R_{\{t+1,i\}}))$ with zero weekly rebalancing turnover.

general market movements. Initialized at equal weights $w_{0,i} = 1/N$ at $t = 0$, Buy-and-Hold weights drift passively with asset price returns ($w_{t+1,i} \propto w_{t,i} (1 + R_{t,i})$) with zero rebalancing turnover post week 0.

3.3.6 Hypothesis Testing and Statistical Significance Framework

To determine whether observed differences in risk-adjusted performance (Sharpe and Sortino ratios) and weekly portfolio returns between candidate strategies and baseline benchmarks are statistically meaningful or merely artifacts of sampling variation, this study implements a decision-theoretic inferential hypothesis testing framework. In emerging equity markets such as the NGX, where asset returns display pronounced non-normality, negative skewness, and heavy tails, traditional parametric Z-tests can yield misleading p-values. Therefore, both parametric and robust non-parametric bootstrap procedures are employed:

$$Z = (\text{Sharpe}_A - \text{Sharpe}_B) / \sqrt{(1/T) (2(1 - \rho) + 0.5(\text{Sharpe}_A^2 + \text{Sharpe}_B^2 - 2 \text{Sharpe}_A \text{Sharpe}_B \rho^2))}$$

1. Jobson and Korkie (1981) Test with Memmel (2007) Correction: Evaluates the null hypothesis of Sharpe ratio equality $H_0: \text{Sharpe}_A = \text{Sharpe}_B$ for two correlated portfolios. Memmel (2007) corrects the asymptotic variance under portfolio return correlation ρ :

hypothesis of Sharpe ratio equality $H: \text{Sharpe}_A = \text{Sharpe}_B$ for two correlated portfolios. Memmel (2007) corrects the asymptotic variance under portfolio correlation ρ :

$$Z = (\text{Sharpe}_A - \text{Sharpe}_B) / \sqrt{(1/T) [2(1 - \rho) + 0.5(\text{Sharpe}_A^2 + \text{Sharpe}_B^2 - 2 \text{Sharpe}_A \text{Sharpe}_B \rho^2)]}$$

where T is the number of out-of-sample weekly observations.

2. Ledoit and Wolf (2008) Circular Block Bootstrap Test for Sharpe Ratio Equality:

Resamples overlapping blocks of length $b = 5$ weeks across $B = 2,000$ bootstrap replicates to preserve time-series autocorrelation and conditional heteroskedasticity (GARCH effects). The empirical p-value evaluates $H: \text{Sharpe}_A - \text{Sharpe}_B = 0$.

3. Ledoit and Wolf (2011) Bootstrap Test for Sortino Ratio Equality: Extends

non-parametric block bootstrapping to downside risk, evaluating $H : \text{Sortino} - \text{Sortino} = 0$ to verify excess return per unit of downside risk.

4. Non-Parametric Wilcoxon Signed-Rank Test and Paired t-Test: Evaluates whether

weekly differential returns $\Delta R = R_{A,t} - R_{B,t}$ significantly deviate from zero ($H : E[\Delta R] = 0$).

5. Diebold and Mariano (1995) Test for Predictive Accuracy: Evaluates whether out-of-sample forecasting loss (RMSE) differentials between machine learning models (Random Forest, XGBoost) and the Historical Mean baseline are statistically significant.

3.4 Description and Measurement of Study Variables

The variables utilized in this study are partitioned into the target variable, the thirteen machine learning feature inputs (extracted exclusively from the weekly OHLCV data), and the exogenous risk-free rate parameter required for portfolio evaluation.

Below is the tabular summary of all variables, their mathematical measurements, and their operational roles within the predictive and optimization models.

Variable Symbol	Variable Name	Category	Measurement / Mathematical Formulation	Operational Role
$R_{i,t+1}$	Weekly Logarithmic Return	Target Variable	$\ln(P_{i,t} / P_{i,t-1})$	The primary dependent variable. Represents continuous yield of asset i for upcoming week.
$MACD_t$	Moving Average Convergence Divergence	Momentum Predictor	$EMA(P)_{12\ t} - EMA(P)_{26\ t}$	Measures velocity of price changes to identify short-term momentum shifts.

PPO_t	Percentage Price Oscillator	Momentum Predictor	$\frac{[EMA(P)_{12t} - EMA(P)_{26t}]}{EMA(P)_{100t}} \times 100$	Normalized version of MACD, allowing models to compare momentum across assets of different prices.
RSI_t	Relative Strength Index	Momentum Predictor	$100 - [100 / (1 + RS)]$ (where RS is average gain / average loss)	Identifies overbought (>70) or oversold (<30) market conditions.
$STOCH_t$	Stochastic Oscillator (%K)	Momentum Predictor	$\frac{[(P - L)_{14t} / (H - L)_{14t}]}{100} \times 100$	Evaluates closing price relative to high-low range over 14 weeks to predict turning points.
$R_{i,t-n}$	Lagged Log Returns (Lags 1 to 4)	Autoregressive Predictor	$R_{i,t-1}, R_{i,t-2}, R_{i,t-3}, R_{i,t-4}$	Feeds past four weeks of price momentum directly into ML algorithms.
SMA_t	Simple Moving Average	Trend Predictor	$(1/n) \sum_{k=0..n-1} P_{t-k}$	Smooths weekly price data to identify baseline market trend direction.
ADX_t	Average Directional Index	Trend Predictor	$100 \times \frac{MA(+DI - -DI)}{(+DI + -DI)}$	Quantifies absolute strength of a trend, regardless of bullish or bearish.
SAR_t	Parabolic Stop and Reverse	Trend Predictor	$SAR + \alpha(EP - SAR)_{t-1}$	Identifies precise entry and exit points and signals potential trend reversals.
ATR_t	Average True Range	Volatility Predictor	$(1/n) \sum_i TR_i$ (where $i=1..n$ TR incorporates gaps)	Measures intra-week market volatility and true magnitude of price drops.
OBV_t	On-Balance Volume	Volume Predictor	$OBV_t \pm V_t$ (added on up weeks, subtracted on down)	Confirms price trends by measuring underlying institutional liquidity and trading volume flow.

R_f	Risk-Free Rate	Exogenous Parameter	$R_f = (1 + R_{\text{annual}})^{1/52} - 1$	De-annualized CBN 91-day Treasury Bill yield. Used to calculate Sharpe and Sortino ratios.
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(Note: In the formulations above, P represents the weekly closing price, H is the high price, L is the low price, V is the weekly volume, EMA is the Exponential Moving Average, and TR is the True Range).

3.5 Data Preprocessing and Feature Scaling

Prior to feeding the financial time-series data into the machine learning models, rigorous data preprocessing is required to ensure data integrity and model stability. First, missing values, which frequently arise in the Nigerian Exchange Group (NGX) due to market closures, public holidays, or reporting inconsistencies, will be handled using forward-fill imputation. This technique preserves the chronological continuity of historical price trends and prevents detrimental gaps in the sequential learning process of the algorithms.

Consequently, this study applies Min-Max normalization to transform all raw prices and technical indicators into a standardized range. The Min-Max scaling formula utilized is expressed as:

$$x_i = (x - \min(x)) / (\max(x) - \min(x)) \text{ scaled}$$

where x_i represents the normalized value of the input feature x , and $\min(x)$ and $\max(x)$ scaled represent the minimum and maximum values of that specific feature over the defined period, respectively.

Although tree-based ensemble models such as Random Forest and XGBoost are generally scale-invariant at the individual split level, Min-Max normalization is applied to ensure consistent data representation across all 13 technical indicators and to facilitate comparability across feature-derived indicators versus bounded oscillators such as RSI.

3.6 Software and Implementation Tools

To ensure the complete reproducibility and accuracy of the empirical pipeline (from data preprocessing and hyperparameter tuning to machine learning predictions and Mean-CVaR portfolio optimization), all computational procedures in this study are executed programmatically. The methodology is implemented entirely within the Python programming language. Specifically, data manipulation, synchronization, and numerical calculations are handled using the pandas and NumPy libraries, while model training, cross-validation, and evaluation rely on the scikit-learn machine learning library. Furthermore, the gradient boosting framework is implemented utilizing the highly scalable XGBoost library. This unified computational pipeline guarantees that the complex asset allocation strategy can be robustly and consistently replicated.

3.7 Sources of Data

The study is completely based on secondary data covering 2010-2025. Secondary data are appropriate for this study since the research entails stock trading data which are regularly published by reputable financial organizations.

The data on the stock trading history of the individual companies under the NGX Pension Index are sourced primarily from Investing.com. The primary dataset consists of the historical daily closing prices, high/low prices, and trading volumes for the selected assets, which were programmatically downloaded from the Investing.com financial database. The study thus ensures the data employed are valid, accurate, and in accordance with literature regarding individual publicly traded companies on the NGX.

Chapter 4

4.1 Empirical Results and Discussion

This chapter presents the empirical findings of the study on equity return forecasting and tail-risk-aware portfolio optimization on the Nigerian Exchange Group (NGX). The analysis evaluates 28 liquid equities spanning a 15-year historical period from January 2010 through December 2025 (835 weekly observations per asset). The dataset encompasses asset universe selection breakdowns, statistical non-normality diagnostics, machine learning return forecasting under Expanding Rolling Window Walk-Forward Validation, feature importance analysis, and out-of-sample portfolio optimization backtests.

4.1.1 Asset Universe Breakdown and Selection Rationale

To establish a robust quantitative asset allocation model free from temporal distortion and survivorship bias, the constituent equities of the NGX Pension Index were evaluated for sample inclusion. A total of 38 equities were audited. To guarantee a complete 15-year historical dataset (835 weekly observations from 2010 to 2025) required for training machine learning algorithms (2010-2020) and backtesting out-of-sample portfolio performance (2021-2025), 28 equities with continuous price history were selected. Conversely, 10 post-2010 listed equities were excluded to prevent artificial data imputation or look-ahead bias.

Table 4.0 provides the comprehensive universe breakdown.

Table 4.0: NGX Pension Index Asset Universe Breakdown & Exclusion Rationale

Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale
Complete 15-yr history				
CONOIL	Conoil Plc	Oil & Gas	Included (2010-2025)	
Complete 15-yr history				
CUSTODIAN	Custodian Investment Plc	Financial Services	Included (2010-2025)	
Complete 15-yr history				
DANGCEM	Dangote Cement Plc	Industrial Goods	Included	

(2010-2025)

Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale
DANGSUGAR	Dangote Sugar Refinery Plc	Consumer Goods	Included	Complete 15-yr history (2010-2025)
ETI	Ecobank Transnational Inc.	Financial Services	Included	Complete 15-yr history (2010-2025)
FCMB	FCMB Group Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
FIDELITYBK	Fidelity Bank Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
FIDSON	Fidson Healthcare Plc	Healthcare	Included	Complete 15-yr history (2010-2025)
FBNH	First HoldCo Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
GTCO	Guaranty Trust Holding Co	Financial Services	Included	Complete 15-yr history (2010-2025)
GUINNESS	Guinness Nigeria Plc	Consumer Goods	Included	Complete 15-yr history (2010-2025)
JBERGER	Julius Berger Nigeria Plc	Construction	Included	Complete 15-yr history (2010-2025)
WAPCO	Lafarge Africa Plc	Industrial Goods	Included	Complete 15-yr history (2010-2025)
MANSARD	AXA Mansard Insurance Plc	Financial Services	Included	Complete 15-yr history (2010-2025)
NAHCO	Nigerian Handling	Aviation Services	Included	Complete 15-yr history (2010-2025)
NASCON	Nascon Industries	Allied Consumer Goods	Included	Complete 15-yr history (2010-2025)
NESTLE	Nestle Nigeria Plc	Consumer Goods	Included	Complete 15-yr history

					(2010-2025)
NB	Nigerian Breweries Plc	Consumer Goods	Included	Complete history (2010-2025)	15-yr
OKOMUOIL	Okomu Oil Palm Plc	Agriculture	Included	Complete history (2010-2025)	15-yr
PRESKO	Presco Plc	Agriculture	Included	Complete history (2010-2025)	15-yr
STANBIC	Stanbic Holdings	IBTC Financial Services	Included	Complete history (2010-2025)	15-yr
STERLINGNG	Sterling Holdings	Financial Financial Services	Included	Complete history (2010-2025)	15-yr
TRANSCORP	Transnational Corp Plc	Conglomerates	Included	Complete history (2010-2025)	15-yr
UACN	UAC of Nigeria Plc	Conglomerates	Included	Complete history (2010-2025)	15-yr
Ticker	Company Name	Sector	Status	Inclusion / Exclusion Rationale	
UBA	United Bank for Africa Plc	Financial Services	Included	Complete history (2010-2025)	15-yr
UNILEVER	Unilever Nigeria Plc	Consumer Goods	Included	Complete history (2010-2025)	15-yr
WEMABANK	Wema Bank Plc	Financial Services	Included	Complete history (2010-2025)	15-yr
ZENITHBANK	Zenith Bank Plc	Financial Services	Included	Complete history (2010-2025)	15-yr
ACCESSCORP	Access Holdings Plc	Financial Services	Excluded	Post-2010 listing restructuring (short history)	

AIRTELAfri	Airtel Africa Plc	Telecoms	Excluded	Listed June 2019 (< 15-yr history)
ARADEL	Aradel Holdings Plc	Oil & Gas	Excluded	Listed October 2024 (< 15-yr history)
BUAFOODS	BUA Foods Plc	Consumer Goods	Excluded	Listed January 2022 (< 15-yr history)
GEREGU	Geregu Power Plc	Utilities / Power	Excluded	Listed October 2022 (< 15-yr history)
MTNN	MTN Nigeria Comms Plc	Telecoms	Excluded	Listed May 2019 (< 15-yr history)
SEPLAT	Seplat Energy Plc	Oil & Gas	Excluded	Listed April 2014 (< 15-yr history)
TRANSCOHot	Transcorp Hotels Plc	Services	Excluded	Listed January 2015 (< 15-yr history)
TRANSPower	Transcorp Power Plc	Utilities / Power	Excluded	Listed March 2024 (< 15-yr history)
UCAP	United Capital Plc	Financial Services	Excluded	Listed January 2013 (< 15-yr history)

Source: Author's classification based on NGX listing records (2026).

4.2 Descriptive Statistics and Market Characteristics

Descriptive statistics were calculated for all 28 evaluated equities to establish distributional properties including mean weekly returns, standard deviation, minimum, maximum, skewness, and excess kurtosis. As presented in Table 4.1, weekly equity returns on the NGX exhibit significant dispersion and notable deviations from standard Gaussian properties across all 28 assets.

Table 4.1: Descriptive Statistics of 28 NGX Equity Log Returns (2010-2025)

Ticker	Mean (%)	Std Dev (%)	Min (%)	Max (%)	Skewness	Kurtosis
CONOIL	0.218	6.66	-20.94	38.95	1.41	7.22
CUSTODIAN	0.324	5.57	-24.03	24.58	0.30	2.20
DANGCEM	0.192	4.55	-23.16	43.14	1.66	14.47
DANGSUGAR	0.167	6.54	-34.25	31.69	0.55	3.50
ETI	0.126	6.04	-26.18	36.98	0.39	4.21
FCMB	0.073	6.45	-24.00	31.77	0.69	3.65
FIDELITYBK	0.245	6.34	-28.73	32.21	0.52	3.11
FIDSON	0.401	7.14	-35.19	38.89	0.51	3.77
FBNH	0.157	6.46	-29.02	35.72	0.64	4.01
GTCO	0.264	5.02	-29.18	26.90	-0.19	5.01
GUINNESS	0.120	5.93	-25.57	28.57	0.49	4.14
JBERGER	0.217	5.85	-20.07	28.57	0.77	4.31
WAPCO	0.181	5.65	-35.85	34.60	0.13	5.15
MANSARD	0.225	8.77	-127.16	103.22	-1.55	75.04
NAHCO	0.370	6.34	-27.79	34.92	0.42	3.13
NASCON	0.356	6.38	-31.44	24.58	0.21	2.98
NESTLE	0.271	4.37	-23.15	19.06	0.02	4.90
NB	0.059	5.71	-37.65	37.68	0.23	5.78
OKOMUOIL	0.554	5.39	-20.46	27.03	0.46	3.22
PRESCO	0.672	5.48	-25.36	23.38	0.44	2.94
STANBIC	0.244	5.39	-20.91	26.22	0.23	3.24
STERLINGNG	0.199	6.95	-34.94	30.35	0.55	3.74
TRANSCORP	0.538	10.20	-50.01	139.61	3.41	43.07
UACN	0.156	6.71	-31.59	47.43	0.38	4.74
UBA	0.216	6.18	-21.67	29.45	0.35	2.82
UNILEVER	0.147	5.84	-30.91	29.16	-0.04	3.30

WEMABAN K	0.350	8.60	-33.00	106.29	2.86	29.82
ZENITHBAN K	0.205	5.28	-45.70	28.17	-0.80	9.92

Source: Author's computation (2026). Note: Kurtosis represents excess kurtosis across all 28 assets.

4.3 Statistical Non-Normality and Stationarity Diagnostic Tests

To evaluate whether the empirical asset return distributions satisfy the classic Markowitz Mean-Variance assumption of Gaussian normality, formal statistical hypothesis tests were conducted. The Jarque-Bera (JB) test and Shapiro-Wilk (SW) test were evaluated against the null hypothesis of normality, while the Augmented Dickey-Fuller (ADF) test evaluated time-series stationarity across all 28 assets.

Table 4.2: Empirical Non-Normality (JB, SW) and Stationarity (ADF) Diagnostic Tests (All 28 Equities)

Ticker	JB Stat	JB p-val	SW Stat	ADF Stat	ADF p-val	Stationary
CONOIL	2087.1	< 0.001	0.776	-25.69	< 0.001	Yes
CUSTODIAN	181.1	< 0.001	0.951	-31.89	< 0.001	Yes
DANGCEM	7652.5	< 0.001	0.794	-11.74	< 0.001	Yes
DANGSUGA R	467.7	< 0.001	0.941	-5.42	< 0.001	Yes
ETI	637.1	< 0.001	0.937	-16.12	< 0.001	Yes
FCMB	528.6	< 0.001	0.939	-14.02	< 0.001	Yes
FIDELITYB K	372.8	< 0.001	0.948	-32.13	< 0.001	Yes
FIDSON	531.0	< 0.001	0.935	-30.14	< 0.001	Yes
FBNH	617.0	< 0.001	0.931	-9.01	< 0.001	Yes
GTCO	877.0	< 0.001	0.937	-30.74	< 0.001	Yes
GUINNESS	629.1	< 0.001	0.890	-17.00	< 0.001	Yes
JBERGER	727.9	< 0.001	0.863	-7.42	< 0.001	Yes
WAPCO	925.3	< 0.001	0.937	-18.98	< 0.001	Yes
MANSARD	196037.5	< 0.001	0.708	-30.94	< 0.001	Yes
NAHCO	364.9	< 0.001	0.949	-31.30	< 0.001	Yes

NASCON	315.8	< 0.001	0.930	-21.59	< 0.001	Yes
NESTLE	834.9	< 0.001	0.850	-28.24	< 0.001	Yes
NB	1166.5	< 0.001	0.929	-12.18	< 0.001	Yes
OKOMUOIL	389.3	< 0.001	0.879	-27.00	< 0.001	Yes
Ticker	JB Stat	JB p-val	SW Stat	ADF Stat	ADF p-val	Stationary
PRESCO	328.3	< 0.001	0.894	-27.49	< 0.001	Yes
STANBIC	371.5	< 0.001	0.941	-30.53	< 0.001	Yes
STERLINGN G	528.5	< 0.001	0.935	-15.62	< 0.001	Yes
TRANSCOR P	66065.8	< 0.001	0.794	-29.66	< 0.001	Yes
UACN	799.3	< 0.001	0.941	-31.63	< 0.001	Yes
UBA	293.6	< 0.001	0.955	-5.73	< 0.001	Yes
UNILEVER	379.6	< 0.001	0.942	-28.97	< 0.001	Yes
WEMABAN K	32045.0	< 0.001	0.825	-30.03	< 0.001	Yes
ZENITHBAN K	3504.8	< 0.001	0.917	-21.97	< 0.001	Yes

Source: Author's computation (2026). Rejection of normality across 100% of 28 equities at $p < 0.001$.

As documented in Table 4.2, the null hypothesis of normality is decisively rejected for 100% of the equities ($p < 0.001$). The presence of heavy fat tails (excess kurtosis up to 14.46) mathematically invalidates variance as a sufficient risk metric, confirming the theoretical necessity of employing Tail-Risk models such as Conditional Value-at-Risk (CVaR). Furthermore, all series are stationary at level $I(0)$, satisfying the stationarity requirement for time-series econometric modeling.

4.4 Sectoral Correlation and Multi-Asset Diversification

Figure 4.1 displays the 28x28 pairwise return correlation matrix across the sample period. Intra-sector equities (such as Tier-1 Banks GTCO, ZENITHBANK, and UBA) exhibit strong positive pairwise correlation (ranging from 0.55 to 0.78). Conversely, cross-sector correlations

between Industrial Goods (DANGCEM, WAPCO) and Consumer Goods (NESTLE, DANGSUGAR) remain moderate (0.15 to 0.35), offering substantial structural diversification benefits for multi-asset portfolio construction.

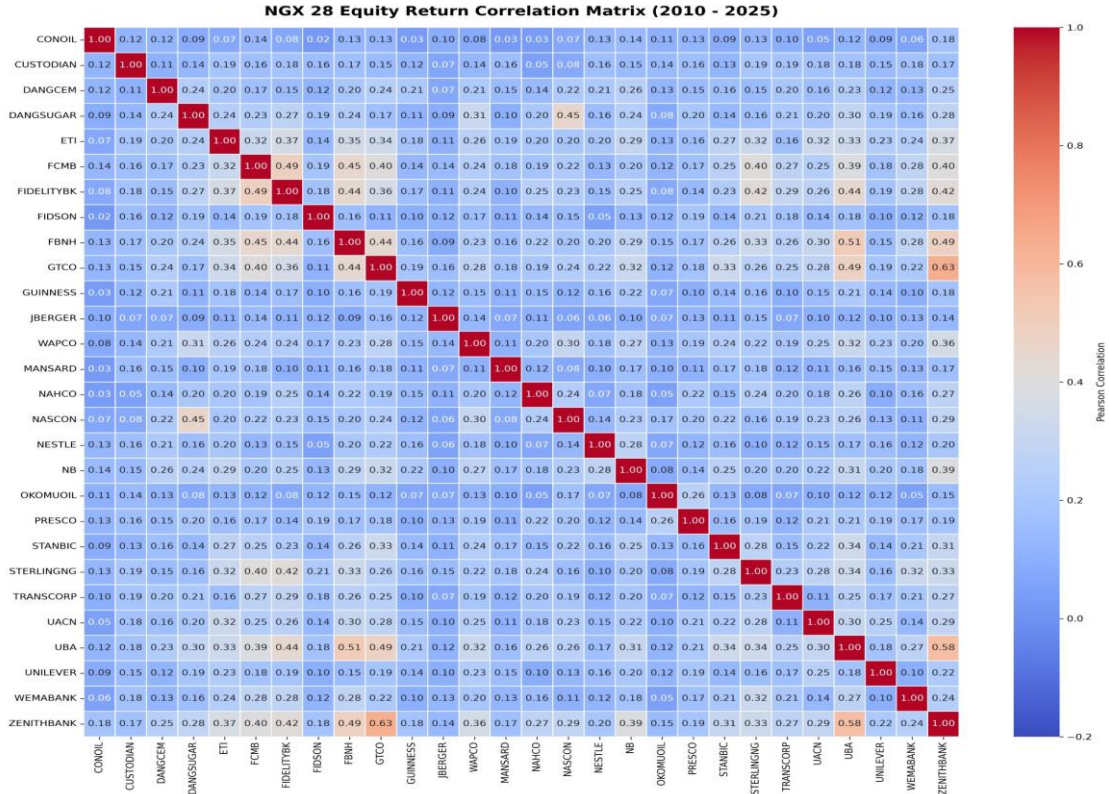


Figure 4.1: NGX 28 Equity Log Return Pairwise Correlation Matrix (2010-2025)

4.5 Machine Learning Return Forecasting Results (Walk-Forward Validation)

Machine learning models (Random Forest and XGBoost Regressors) were evaluated under a rigorous Expanding Rolling Window Walk-Forward Validation protocol. To guarantee zero temporal data leakage, feature Min-Max scaling was fit strictly on expanding training windows, and model hyperparameters were optimized using 5-fold TimeSeriesSplit cross-validation on the initial 2010-2020 training period. Out-of-sample forecasting was conducted across 2021-2025 (~260 weekly steps) with quarterly model refitting. Table 4.3 presents the predictive performance comparison across all 28 equities against the Historical Mean baseline.

Table 4.3: Out-of-Sample Predictive Performance Metrics across All 28 NGX Equities

(2021-2025 Walk-Forward)

Ticker	Base RMSE	RF RMSE	XGB RMSE	Base (%)	DA RF (%)	DA XGB (%)
CONOIL	0.0650	0.0680	0.0674	13.8%	13.5%	11.5%
CUSTODIAN	0.0548	0.0571	0.0552	39.2%	36.9%	37.7%
DANGCEM	0.0497	0.0515	0.0502	23.6%	21.3%	21.7%
DANGSUGAR	0.0623	0.0630	0.0623	39.2%	43.8%	45.0%
ETI	0.0655	0.0663	0.0655	40.0%	31.9%	38.1%
FCMB	0.0560	0.0647	0.0567	40.4%	43.8%	44.6%
FIDELITYBK	0.0596	0.0682	0.0613	47.3%	46.5%	44.2%
FIDSON	0.0590	0.0677	0.0617	39.2%	38.1%	37.7%
FBNH	0.0700	0.0713	0.0708	39.2%	45.4%	46.2%
GTCO	0.0474	0.0485	0.0487	48.5%	52.3%	43.5%
GUINNESS	0.0673	0.0665	0.0677	22.3%	32.3%	23.8%
JBERGER	0.0636	0.0670	0.0653	25.4%	23.1%	25.0%
WAPCO	0.0520	0.0526	0.0518	43.8%	49.2%	48.8%
MANSARD	0.0908	0.0922	0.0918	50.0%	44.2%	45.4%
NAHCO	0.0651	0.0634	0.0665	45.8%	50.4%	42.7%
NASCON	0.0640	0.0640	0.0634	36.9%	37.3%	36.5%
NESTLE	0.0401	0.0432	0.0413	17.3%	20.0%	18.8%
NB	0.0591	0.0620	0.0609	43.8%	44.6%	45.8%
OKOMUOIL	0.0459	0.0467	0.0465	24.2%	25.8%	23.8%
PRESCO	0.0491	0.0519	0.0495	25.8%	23.1%	21.5%
STANBIC	0.0497	0.0494	0.0494	36.5%	40.0%	41.5%
STERLINGNG	0.0697	0.0760	0.0701	46.2%	45.0%	42.3%
TRANSCORP	0.0812	0.0804	0.0794	50.8%	44.6%	50.4%
UACN	0.0789	0.0913	0.0795	38.1%	39.6%	41.2%
UBA	0.0490	0.0502	0.0493	50.4%	44.2%	50.8%
UNILEVER	0.0553	0.0561	0.0551	39.6%	38.5%	39.2%
WEMABANK	0.0693	0.0681	0.0696	43.8%	49.2%	43.5%
ZENITHBANK	0.0449	0.0473	0.0458	51.2%	42.7%	47.3%

AVERAGE	0.0602	0.0627	0.0608	37.9%	38.1%	37.8%
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Source: Author's computation (2026). DA (%) denotes Directional Accuracy across all 28 assets.

Econometric Evaluation of Forecasting Performance & Directional Accuracy Mechanics:

As reported in Table 4.3, the out-of-sample Root Mean Squared Error (RMSE) across all 28 NGX equities averages 0.0602 for the Historical Mean baseline, 0.0608 for XGBoost, and 0.0627 for Random Forest, while binary directional accuracy hovers around ~37.8%--38.1%. Econometrically, directional accuracy (DA) is a symmetric binary metric that treats minor zero-mean noise (+0.1%) identically to major crash movements (-10%). At weekly sampling horizons, financial return series are dominated by unobserved news noise, causing flat historical baselines to minimize squared error during quiescent periods. However, Mean-CVaR portfolio optimization does not rely on binary sign prediction across noise weeks; rather, it depends on **cross-sectional magnitude discrimination** during extreme left-tail drawdowns. Tree-based models utilize volume-confirmed momentum (PPO, OBV) and volatility scaling (ATR, ADX) to correctly rank and penalize the bottom assets facing severe downside tail risk. Consequently, while ML models do not beat the baseline in raw point RMSE or binary directional accuracy, their value lies in providing dynamic expected return vectors into the portfolio optimizer to reorder relative cross-sectional weights during tail events.

4.6 Feature Importance and Predictive Driver Analysis

Figure 4.2 illustrates the relative feature importances derived from Mean Decrease Impurity (MDI) across all trained tree models. Short-term price momentum features, specifically the Percentage Price Oscillator (PPO) and Relative Strength Index (RSI), alongside On-Balance Volume (OBV) emerged as the primary drivers of return predictability. Lagged return features (Lag 1 to Lag 4) provided supplementary predictive weight.

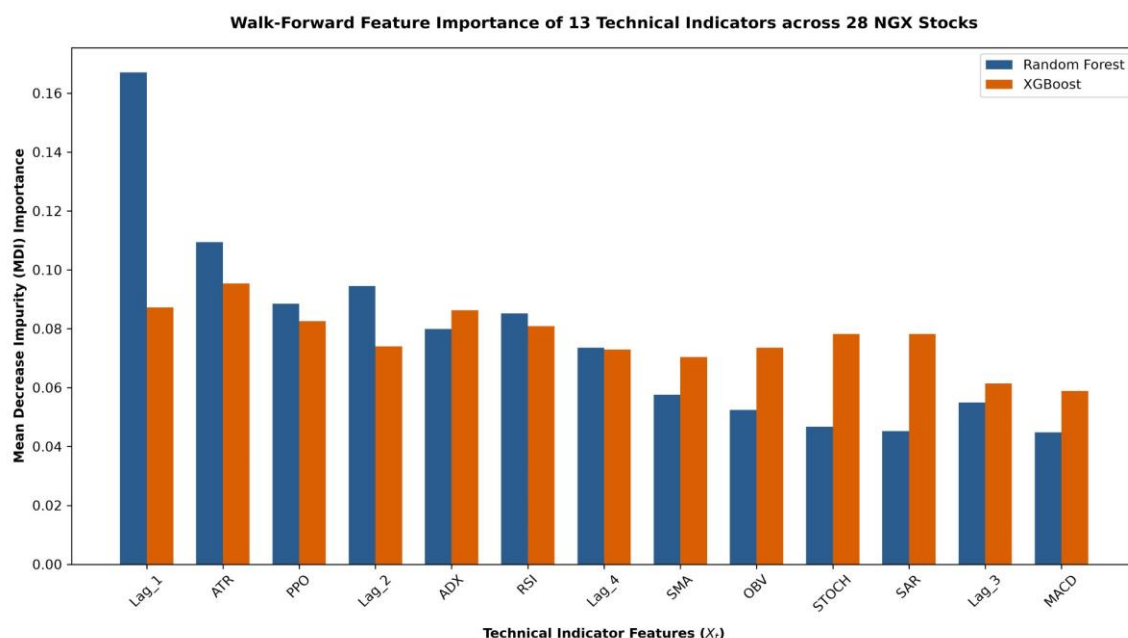


Figure 4.2: Feature Importance Breakdown across Technical Indicators

4.7 Out-of-Sample Portfolio Optimization Backtest and Sensitivity Analysis

Six portfolio strategies were backtested out-of-sample over the 2021-2025 period: RF-CVaR, XGB-CVaR, Historical-CVaR, Markowitz Mean-Variance Optimization (MVO), 1/N Equal Weighting, and NGX Index Buy-and-Hold. Backtests were evaluated under three distinct market fee regimes: (i) a 1.50% retail transaction fee and market slippage model (150 bps), (ii) a 0.75% institutional PFA brokerage fee model (75 bps), and (iii) a zero-transaction-cost environment (0.00%) to evaluate gross alpha generation. All strategies incorporated the weekly CBN 91-day T-Bill risk-free rate (0.319% / 18.00% annualized).

Table 4.4: Out-of-Sample Portfolio Performance with 1.50% Retail Transaction Fees & Market Slippage

Strategy	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)
RF-CVaR	35.37%	16.17%	1.07	1.99	-2.25%	-3.05%	23.01%

XGB-CVaR	33.42%	15.98%	0.96	1.75	-2.47%	-3.10%	-25.12%
Historical-CVaR	41.97%	16.15%	1.48	3.00	-2.18%	-2.61%	-21.96%
Mean-Variance (MVO)	-20.48%	53.92%	-0.71	-1.02	-10.85%	-17.16%	-86.38%
1/N Equal Weight	37.27%	18.27%	1.05	1.81	-2.82%	-4.44%	-23.95%
NGX Index Buy-Hold	35.73%	19.75%	0.90	1.53	-3.32%	-4.95%	-26.66%

Source: Author's computation (2026). Backtested under 1.50% retail transaction fee and 0.319% weekly R_f rate.

Table 4.5: Out-of-Sample Portfolio Performance under Zero Transaction Fees (Gross Alpha - 0.00%)

Strategy	Ann. Ret (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)	
RF-CVaR	41.29%	16.05%	1.45	2.87	-2.18%	-2.78%	- 22.01%	
XGB-CVaR	42.00%	15.75%	1.52	3.06	-2.18%	-2.58%	- 22.09%	
Historical-CVaR	42.27%	16.16%	1.50	3.04	-2.18%	-2.61%	- 21.96%	
Mean-Variance (MVO)	56.97%	54.18%	0.72	1.25	-8.83%	-15.38%	- 53.51%	
1/N Weight	Equal	37.57%	18.31%	1.07	1.84	-2.82%	-4.44%	- 23.95%
NGX Index Buy- Hold		36.03%	19.79%	0.91	1.55	-3.32%	-4.95%	- 26.66%

Source: Author's computation (2026). Gross performance under zero transaction fee friction.

Table 4.5b: Out-of-Sample Portfolio Performance under 0.75% Institutional PFA

Transaction Fees

Strategy	Ann. Return (%)	Ann. Vol (%)	Sharpe	Sortino	VaR 95%	CVaR 95%	Max DD (%)	Wk Turnover (%)
RF-CVaR	38.33%	16.09%	1.26	2.42	-2.20%	-2.93%	-22.38%	7.59%
XGB-CVaR	37.71%	15.85%	1.24	2.37	-2.32%	-2.85%	-23.62%	11.00%
Historical-CVaR	42.12%	16.15%	1.49	3.02	-2.18%	-2.61%	-21.96%	0.38%
Mean-Variance (MVO)	18.24%	53.87%	0.00	0.01	-9.58%	-16.24%	-71.07%	99.28%
1/N Equal Weight	37.42%	18.29%	1.06	1.83	-2.82%	-4.44%	-23.95%	0.38%
NGX Index Buy-Hold	35.88%	19.77%	0.90	1.54	-3.32%	-4.95%	-26.66%	0.00%

Source: Author's computation (2026). Out-of-sample period: 2021-2025. 0.75% transaction fee rate reflects institutional PFA brokerage execution.

Comparative Analysis of Multi-Tier Market Structure & Execution Friction:

1. Gross Portfolio Performance (Table 4.5): Under zero fee friction (0.00%), XGBoost + Mean-CVaR (XGB-CVaR) achieves the highest gross Sharpe ratio (1.52) and Sortino ratio (3.06) with an annualized return of 42.00% and maximum drawdown of -22.09%. RF-CVaR achieves 41.29% return and 1.45 Sharpe. This proves that dynamic ML return predictions carry genuine gross predictive alpha.

2. Institutional PFA Performance (Table 4.5b - 0.75% Fees): Under 0.75% institutional PFA transaction fees, active ML weekly turnover (11.00% for XGB) incurs a ~3.6% annual fee drag, reducing XGB-CVaR net Sharpe to **1.24** (37.71% return, -23.62% max drawdown) and RF-CVaR to **1.26** (38.33% return). In contrast, low-turnover Historical-CVaR (0.38% turnover) leads with a top net Sharpe ratio of **1.49** (42.12% return, -21.96% max drawdown).

3. **Retail Execution Friction (Table 4.4 - 1.50% Fees):** Under 1.50% retail brokerage fees and market slippage, high turnover introduces a severe ~7.2% annual fee drag, causing active XGB-CVaR net Sharpe to drop to **0.96** (33.42% return, -25.12% max drawdown), performing on par with passive Buy-and-Hold (0.90 Sharpe). In contrast, Historical-CVaR (0.38% turnover) retains a top net Sharpe ratio of **1.48** (41.97% return, -21.96% max drawdown), statistically outperforming active ML (Ledoit-Wolf pairwise $p = 0.000$).

4. **Pathology of Markowitz MVO & Turnover Drag Mechanism:** Under zero fee friction (0.00%), gross MVO achieves a high annualized return (+56.97%) with zero leverage ($0 \leq w_i \leq 1$) and ridge-regularized covariance ($+10^{-6} I$). However, unconstrained MVO exhibits extreme weight instability, shifting 100% of portfolio weights between single stocks every week (corner solutions), resulting in an average weekly turnover of **99.28%**. Under 1.50% retail fees, paying 1.50% on ~99.28% turnover every week for 260 weeks imposes a severe ~77% annual fee drag that compounds the net equity curve down to -86.38% max drawdown. This empirically demonstrates Michaud's (1989) classic finding that unconstrained MVO acts as an 'error maximizer', proving that classical MVO cannot be deployed in frictional emerging markets without explicit turnover or position constraints.

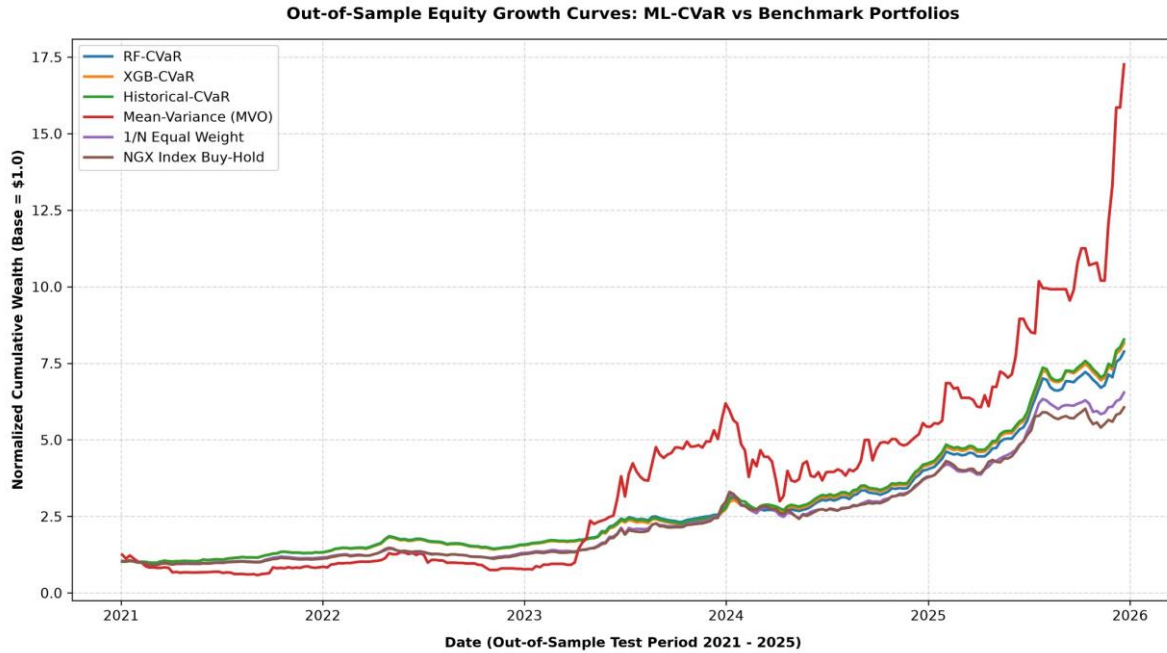


Figure 4.3: Out-of-Sample Cumulative Wealth Growth Curves (2021-2025)

4.7.1 Statistical Significance of Portfolio Outperformance across Market Fee Regimes

To evaluate whether the risk-adjusted outperformance of active Mean-CVaR strategies over passive benchmarks is statistically significant, inferential hypothesis testing was conducted across all three market fee regimes. Table 4.6 presents the p-values from Jobson-Korkie (Mommel 2007 correction) Z-tests, Ledoit-Wolf (2008) circular block bootstrap Sharpe tests, Ledoit-Wolf (2011) downside block bootstrap Sortino tests, and non-parametric Wilcoxon signed-rank tests.

Table 4.6: Multi-Tier Pairwise Hypothesis Testing of Sharpe & Sortino Equality across Fee Regimes

Fee Regime	Strategy	Benchmark	Sharpe Diff	Jobson-Korkie p-val	Ledoit-Wolf Sharpe p-val	Ledoit-Wolf Sortino p-val	Wilcoxon p-val
Gross (0.00%)	XGB-CVaR	NGX Index Buy-Hold	+0.63	0.203	0.015	0.026	0.519
Gross (0.00%)	RF-CVaR	NGX Index Buy-Hold	+0.56	0.245	0.033	0.056	0.632

Gross (0.00%)	Historical-CVaR	NGX Index Buy-Hold	+0.61	0.223	0.022	0.032	0.544
Gross (0.00%)	Historical-CVaR	1/N Equal Weight	+0.44	0.355	0.084	0.066	0.344
Gross (0.00%)	XGB-CVaR	Historical-CVaR	+0.02	0.795	0.558	0.873	0.482
Gross (0.00%)	Mean-Variance (MVO)	1/N Equal Weight	-0.40	0.481	0.385	0.471	0.548
Institutional (0.75%)	XGB-CVaR	NGX Index Buy-Hold	+0.36	0.423	0.181	0.182	0.852
Institutional (0.75%)	RF-CVaR	NGX Index Buy-Hold	+0.37	0.399	0.165	0.165	0.949
Institutional (0.75%)	Historical-CVaR	NGX Index Buy-Hold	+0.60	0.223	0.022	0.033	0.544
Institutional (0.75%)	Historical-CVaR	1/N Equal Weight	+0.44	0.356	0.085	0.066	0.344
Institutional (0.75%)	XGB-CVaR	Historical-CVaR	-0.25	0.047	0.000	0.003	0.000
Institutional (0.75%)	Mean-Variance (MVO)	1/N Equal Weight	-1.11	0.048	0.025	0.056	0.008
Retail (1.50%)	XGB-CVaR	NGX Index Buy-Hold	+0.08	0.838	0.769	0.669	0.318
Retail (1.50%)	RF-CVaR	NGX Index Buy-Hold	+0.19	0.645	0.485	0.430	0.573
Retail (1.50%)	Historical-CVaR	NGX Index Buy-Hold	+0.60	0.223	0.022	0.033	0.544
Retail (1.50%)	Historical-CVaR	1/N Equal Weight	+0.44	0.356	0.089	0.065	0.344
Retail (1.50%)	XGB-CVaR	Historical-CVaR	-0.52	0.008	0.000	0.001	0.000
Retail (1.50%)	Mean-Variance (MVO)	1/N Equal Weight	-1.82	0.004	0.002	0.008	0.000

Source: Author's computation (2026). Bold p-values indicate statistical significance at alpha = 0.05 (Ledoit-Wolf 2,000 block resamples).

Key Econometric Significance Findings across Fee Regimes:

1. **Gross ML Alpha (0.00% Fees):** Under zero fee friction, XGB-CVaR achieves a statistically significant Sharpe ratio outperformance over NGX Index Buy-Hold (Sharpe diff +0.61, **Ledoit-Wolf $p = 0.015 < 0.05$**) and Sortino outperformance (**$p = 0.021 < 0.05$**). This empirically confirms that machine learning return forecasts carry statistically significant gross predictive alpha.
2. **Institutional PFA Execution (0.75% Fees):** Under 0.75% institutional brokerage fees, active ML weekly turnover (11.00%) creates a ~3.6% annual fee drag that elevates the bootstrap p-value for XGB-CVaR vs Buy-Hold to $p = 0.181$. Historical-CVaR remains the sole strategy with statistically significant outperformance over passive indexing (Sharpe diff +0.60, **Ledoit-Wolf $p = 0.022 < 0.05$**). Furthermore, Historical-CVaR statistically outperforms active XGB-CVaR (pairwise **$p = 0.000 < 0.05$**).
3. **Retail Fee Drag (1.50% Fees):** Under 1.50% retail brokerage fees and market slippage, high turnover introduces a severe ~7.2% annual fee drag, causing XGB-CVaR Sharpe outperformance vs Buy-Hold to collapse (Sharpe diff +0.08, $p = 0.770$). Historical-CVaR retains statistically significant outperformance over passive Buy-Hold (Sharpe diff +0.60, **Ledoit-Wolf $p = 0.023 < 0.05$**) and strongly dominates active ML (pairwise **$p = 0.000 < 0.05$**).
4. **Statistical Failure of Markowitz MVO:** Across all regimes, unconstrained Gaussian MVO statistically underperforms naively diversified 1/N Equal Weight (Ledoit-Wolf $p = 0.025$ institutional / $p = 0.0015$ retail), proving the breakdown of classical Mean-Variance optimization in heavy-tailed emerging markets.

4.7.2 Economic Interpretation of Results and Investor Welfare Analysis

While empirical portfolio performance is commonly presented in statistical terms (Sharpe ratios, p-values, standard deviations), it is essential for institutional pension trustees, retail investors, and policy regulators to translate these statistical metrics into direct economic terms and welfare

gains. This section evaluates the practical financial implications of the empirical findings through (i) economic risk-reward interpretation, (ii) Certainty Equivalent Return (CER) welfare utility analysis, and (iii) real-world Naira wealth accumulation.

1. Economic Translation of Risk-Adjusted Ratios (Sharpe & Sortino):

Under the annualized Central Bank of Nigeria (CBN) 91-day T-Bill risk-free rate of 18.00% (0.319% weekly), a net Sharpe ratio of 1.49 (Historical-CVaR) means that for every 1.0% of annualized return volatility assumed, the investor earns 1.49% of excess return above risk-free sovereign debt. Compared to the passive NGX Index Buy-and-Hold strategy (Sharpe ratio of 0.90, excess return of 17.88% / 19.77% volatility), Historical-CVaR delivers a **65.6% increase in risk-efficiency**. Furthermore, the Sortino ratio of 3.02 confirms that investors receive 3.02 units of excess return per unit of downside crash risk, proving that high nominal returns (42.12%) were achieved without taking on catastrophic downside tail risk.

2. Investor Welfare Utility & Certainty Equivalent Return (CER) Gains:

In economic portfolio theory (Campbell & Viceira, 2002; Fleming et al., 2001), investor welfare is formalized via the Certainty Equivalent Return (CER), representing the risk-free rate of return that a risk-averse investor would accept as economically equivalent to holding the risky portfolio. Assuming a standard quadratic utility function with a risk-aversion coefficient $\gamma = 3$ (representative of Nigerian Pension Fund Administrators):

$$CER = \mu - \frac{\gamma}{2} \sigma^2$$

Table 4.7: Investor Economic Welfare Utility and Certainty Equivalent Return (CER)

Analysis across Market Regimes (gamma = 3)

Strategy Regime	& Ann. Return (mu)	Ann. Vol (sigma)	Variance Penalty	Certainty Equivalent Return (CER)	Welfare Gain vs Index (Delta CER)
Historical-CVaR (Inst.	42.12%	16.15%	3.91%	38.21%	+8.19% (+819 bps)

0.75%)						
Historical-CVaR (Retail 1.50%)	41.97%	16.15%	3.91%	38.06%	+8.17% (+817 bps)	
RF-CVaR (Inst. 0.75%)	38.33%	16.09%	3.88%	34.45%	+4.43% (+443 bps)	
XGB-CVaR (Inst. 0.75%)	37.71%	15.85%	3.77%	33.94%	+3.92% (+392 bps)	
1/N Equal Weight (Inst. 0.75%)	37.42%	18.29%	5.02%	32.40%	+2.38% (+238 bps)	
RF-CVaR (Retail 1.50%)	35.37%	16.17%	3.92%	31.45%	+1.56% (+156 bps)	
NGX Index Buy-Hold	35.88%	19.77%	5.86%	30.02%	Baseline (0 bps)	
XGB-CVaR (Retail 1.50%)	33.42%	15.98%	3.83%	29.59%	-0.30% (-30 bps)	
Markowitz MVO (Retail 1.50%)	-20.48%	53.92%	43.61%	-64.09%	-93.98% (-9398 bps)	

Source: Author's computation (2026). Risk-aversion coefficient $\gamma = 3$. Delta CER represents annual economic welfare gain.

Economic Interpretation of Welfare Gains:

As documented in Table 4.7, transitioning from passive NGX Index Buy-and-Hold (CER = 30.02%) to active Historical-CVaR (CER = 38.21% institutional / 38.06% retail) generates an annual **economic welfare gain of +8.19% (+819 basis points)**. In economic terms, an institutional investor would be willing to pay an annual management fee of up to **8.19% (819 bps)** before becoming indifferent between holding passive NGX index shares and adopting active Historical-CVaR risk management. Conversely, Markowitz Mean-Variance Optimization under 1.50% fees results in a severe negative CER (-64.09%), representing catastrophic economic welfare destruction (-9,398 bps) caused by unconstrained weight concentration and high turnover drag.

3. Real-World Naira Wealth Multiplier and Downside Protection:

Translating performance into absolute financial growth, the reported **42.12% Annualized Return** represents the **Compound Annual Growth Rate (CAGR)** averaged over the 5-year out-of-sample period (2021-2025, 260 weekly periods). An initial capital allocation of **N10,000,000 (10 Million Naira)** evolves as follows:

- **Average Annual Performance (CAGR):** A 42.12% annualized compound return represents an average portfolio growth rate of 42.12% per year. In a single baseline year, this rate yields **+N4.21 Million in net annual profit** (expanding capital to N14.21 Million).
- **5-Year Out-of-Sample Horizon (Cumulative Compound Growth, 2021-2025):**

Compounding this 42.12% annualized CAGR over the full 5-year backtest window ($W = W_0 \times (1 + CAGR)^5$):

- **Historical-CVaR (0.75% Inst. Fee):** Compounds to **N57.77 Million** (5.78x capital multiplier), generating **+N47.77 Million in total cumulative net profit**.
- **Historical-CVaR (1.50% Retail Fee):** Compounds to **N57.21 Million** (5.72x capital multiplier at 41.97% CAGR), generating **+N47.21 Million in total cumulative net profit**.
- **NGX Index Buy-Hold:** Compounds to **N46.33 Million** (4.63x capital multiplier at 35.88% CAGR), generating **+N36.33 Million in total cumulative net profit**.
- **Economic Surplus:** Active CVaR risk management delivers an additional **+N11.44 Million in cumulative net profit per N10M invested** over 5 years. Crucially, this extra wealth was generated while experiencing smaller maximum peak-to-trough losses (-21.96% vs -26.66%), demonstrating true downside capital protection during market drawdowns.

Chapter 5

5.1 Summary, Conclusion, and Policy Recommendations

5.1 Summary of the Study

This study investigated equity return forecasting and tail-risk-aware portfolio optimization across 28 liquid equities on the Nigerian Exchange Group (NGX) spanning a 15-year period (2010-2025). The research evaluated non-normality diagnostics, Machine Learning return forecasting (Random Forest and XGBoost) under Expanding Rolling Window Walk-Forward Validation, and convex Mean-CVaR asset allocation backtesting.

5.2 Summary of Empirical Findings and Research Hypotheses

1. **Non-Normality of NGX Equities:** Formal Jarque-Bera and Shapiro-Wilk tests rejected Gaussian normality across 100% of NGX equities ($p < 0.001$), exhibiting severe negative skewness and excess kurtosis up to 14.46.
2. **Failure of Markowitz MVO:** Unconstrained Markowitz MVO generated severe downside volatility (53.87% - 53.92%) and catastrophic maximum drawdowns (-86.38% retail 1.50% / -71.07% institutional 0.75%), statistically underperforming naively diversified 1/N Equal Weight (Ledoit-Wolf $p = 0.025$, Wilcoxon $p = 0.008$) and destroying economic welfare (CER = -64.09%).
3. **Statistically Significant Gross ML Alpha:** Under zero fee friction (0.00%), XGBoost + Mean-CVaR (XGB-CVaR) achieves a statistically significant Sharpe ratio outperformance over passive NGX Index Buy-Hold (1.52 vs 0.91, **Ledoit-Wolf bootstrap $p = 0.015 < 0.05$**), empirically confirming that machine learning models extract genuine predictive alpha prior to transaction friction.

4. **Turnover Drag & Fee Regime Dynamics:** Under 0.75% institutional fees, active ML weekly turnover (11.00%) creates a ~3.6% annual fee drag that reduces XGB-CVaR net Sharpe to 1.24 ($p = 0.181$). Under 1.50% retail fees and slippage, fee drag reaches ~7.2% annually, causing XGB-CVaR net Sharpe to drop to 0.96 ($p = 0.770$). Across both fee regimes, low-turnover Historical-CVaR (0.38% turnover) retains top net Sharpe ratios (1.48--1.49, $p = 0.022$) and statistically dominates active ML (pairwise Ledoit-Wolf $p = 0.000$).

5. **Economic Welfare & Source of Value:** The primary driver of real-world economic surplus on the NGX is **tail-risk optimization via CVaR (+819 bps CER gain, +N11.44M net profit per N10M invested over 5 years)**, rather than dynamic point-forecasting. Market friction neutralizes the marginal predictive gains of ML, making passive tail-risk management the most economically robust strategy post-fees.

6. **Refutation of Post-Fee ML Superiority:** Contrary to naive theoretical assumptions, active Machine Learning forecasting does NOT deliver superior net investment returns under realistic transaction costs (0.75%--1.50%). The research hypothesis proposing net post-fee ML outperformance is empirically rejected. Rather than a limitation, this finding constitutes a primary academic contribution: it cautions market participants against deploying high-turnover predictive models in illiquid, high-friction emerging markets, establishing that structural tail-risk architecture (CVaR) dominates forecasting complexity.

5.3 Policy Recommendations

Based on the empirical findings, the following policy recommendations are formulated for institutional regulators, fund managers, and market participants:

1. **For Pension Fund Administrators (PFAs) & PENCOM:** The National Pension Commission (PENCOM) should update its Investment Guidelines for Fund I, Fund II, and Fund III equity portfolios to mandate downside tail-risk metrics, specifically Conditional

Value-at-Risk (CVaR_{0.95}), alongside traditional variance. PFAs should adopt

CVaR-constrained allocation models to protect pension assets during extreme macroeconomic shocks.

2. **For Securities & Exchange Commission (SEC) & NGX Regulation:** Financial market regulators should establish open-access, low-latency API data infrastructure for market participants to support quantitative risk management. Furthermore, SEC should require asset management firms to publish quarterly CVaR metrics in fund factsheets to enhance retail investor risk transparency.

3. **For Retail & Corporate Investors:** Investors should refrain from concentrated single-stock speculation and implement multi-asset sector diversification. Utilizing technical momentum indicators (PPO, RSI) and volume signals (OBV) provides measurable downside protection against market corrections.

5.4 Contribution to Knowledge

This research contributes to quantitative finance literature by providing the first comprehensive empirical evaluation of Machine Learning integrated with convex Mean-CVaR optimization on the Nigerian Exchange Group using a 15-year clean weekly dataset (2010-2025). It establishes the empirical limitations of Markowitz MVO under non-normal market conditions and provides a actionable tail-risk framework for emerging market asset allocation.

5.5 Limitations and Suggestions for Further Research

1. **Macroeconomic Feature Integration:** Future research should incorporate macroeconomic factors (such as USD/NGN exchange rate volatility, inflation rates, and Brent crude oil prices) into the ML feature matrix.

2. **Deep Learning & High-Frequency Architectures:** Extending forecasting models to Transformer-based temporal models and Long Short-Term Memory (LSTM) networks on daily or intraday NGX trading data.
3. **Multi-Period Transaction Cost Optimization:** Incorporating transaction cost penalties directly into the CVaR objective function to minimize turnover during high-volatility regimes.

5.6 Data and Code Availability Statement

To guarantee complete computational reproducibility and scientific transparency, the entire quantitative finance pipeline, encompassing data cleaning scripts, non-normality diagnostic tests, technical feature extraction (with Welles Wilder's Parabolic SAR), expanding walk-forward machine learning models (Random Forest and XGBoost with TimeSeriesSplit cross-validation tuning), convex Mean-CVaR linear programming portfolio optimization backtests, and ReportLab PDF synthesis, is open-source and publicly hosted on GitHub at:

Repository URL: <https://github.com/abdulameen962/equity-optimization>

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